Universal Connections and a Characteristic Class for Euclidean Space

By

Roger Sherwood Schlafly

B.S.E. (Princeton University) 1976
M.A. (University of California) 1977

DISSERTATION

Submitted in partial satisfaction of the requirements for the degree of

DOCTOR OF PHILOSOPHY

in

Mathematics

in the

GRADUATE DIVISION

OF THE

UNIVERSITY OF CALIFORNIA, BERKELEY

Approved:

I M Singer	5/21/80
Chairman	Date
S. S. Chern	4/21/80
Eugene Wigner	5/12/80

Universal Connections and a Characteristic Class for Euclidean Space

by Roger Schlafly

Department of Mathematics

Abstract

Let BG be the classifying space of a compact Lie group G. In chapter I, I give a new proof of the theorem of Narasimhan and Ramanan that the canonical connection on the universal bundle over BG is universal. The proof also yields real-analytic and equivariant versions of this theorem. I also give a uniqueness theorem.

BG is approximated by Grassmannians, and in chapter II I consider the problem of finding the lowest dimension Grassmannian for which the Narasimhan-Ramanan theorem applies. A method is given for solving this problem locally, and this is explicitly worked out in several cases. Some further remarks appear in chapter III.

Let F be the curvature of some connection on some principal bundle on $\mathbb{R}^4$. In chapter IV, I show that if F decays as fast as $O(1/(r^2 \log r))$ as r tends to infinity, then $\frac{1}{8\pi^2}\int \operatorname{tr} F^2$ is an integer. If F decays like $O(1/r^2)$, any real value is possible. There is an analogous statement for $\mathbb{R}^{2n}$ $(n > 2)$, although it fails for $\mathbb{R}^2$. An alternative approach to this theorem is outlined in chapter V.

Universal Connections and a Characteristic Class for Euclidean Space

by Roger Schlafly

Department of Mathematics

Abstract

Let BG be the classifying space of a compact Lie group G. In chapter I, I give a new proof of the theorem of Narasimhan and Ramanan that the canonical connection on the universal bundle over BG is universal. The proof also yields real-analytic and equivariant versions of this theorem. I also give a uniqueness theorem.

BG is approximated by Grassmannians, and in chapter II I consider the problem of finding the lowest dimension Grassmannian for which the Narasimhan-Ramanan theorem applies. A method is given for solving this problem locally, and this is explicitly worked out in several cases. Some further remarks appear in chapter III.

Let F be the curvature of some connection on some principal bundle on $\mathbb{R}^4$. In chapter IV, I show that if F decays as fast as $O(1/(r^2 \log r))$ as r tends to infinity, then $\frac{1}{8\pi^2}\int \operatorname{tr} F^2$ is an integer. If F decays like $O(1/r^2)$, any real value is possible. There is an analogous statement for $\mathbb{R}^{2n}$ $(n > 2)$, although it fails for $\mathbb{R}^2$. An alternative approach to this theorem is outlined in chapter V.

Universal Connections and a Characteristic Class for Euclidean Space

Copyright © 1980

by Roger Schlafly

Acknowledgment

I would like to thank my advisor, Iz Singer, for advice and encouragement. Without his help, this thesis would not be possible. I would also like to thank everyone else who was influential in my graduate education, particularly, S. S. Chern, Rob Kirby, and Jack Wagoner.

Universal Connections and a Characteristic Class for Euclidean Space

by Roger Schlafly

Table of Contents

Chapter I. Universal Connections: The Narasimhan-Ramanan Theorem 1
1. The Orthogonal Group 1
2. The Narasimhan-Ramanan Theorem 4
3. An Equivariant Version 6
4. A Uniqueness Theorem 7
References 11

Chapter II. Universal Connections: The Local Problem 12
1. Introduction 12
2. Definition of the Universal Connection 13
3. Cartan's Method 15
4. O(n)-Bundles over $\mathbb{R}^2$ 18
5. SU(2) and U(2) Bundles over $\mathbb{R}^2$ 19
6. O(2) and O(3) Bundles over $\mathbb{R}^3$ 22
7. O(2) Bundles over $\mathbb{R}^4$ 28

Chapter III. Universal Connections: Further Remarks 32
8. The Quaternionic Grassmanian 32
9. The Universal Connection for Families 35
References for Chapters II and III 37

Chapter IV. A Chern Number for Gauge Fields over $\mathbb{R}^4$ 38
1. Introduction 38
2. Outline of the Proof 39
3. The Holonomy Propagator 41
4. Estimates in the Radial Gauge 44
5. The Holonomy at Infinity 47
6. An Example 54
7. A Generalization 55
References 58

Chapter V. Convergence of an Index Problem 59
1. The Dirac Operator 59
2. The Eta Invariant 60
3. Further Remarks 62
References 65

Chapter I

UNIVERSAL CONNECTIONS

by Roger Schlafly*

Let H be a compact Lie group. It is well-known [5] that principal H-bundles over a manifold M are classified by homotopy classes of maps into a certain space BH called the classifying space. In particular, every principal H-bundle over M is the pull-back of the canonical bundle over BH by some map from M into BH. In [4], Narasimhan and Ramanan showed that the canonical bundle over BH has a connection which is universal in the sense that any connection on any principal H-bundle over M may be induced by some map from M to BH.

A simple proof of the Narasimhan-Ramanan theorem will be given in this paper. The method yields the following generalizations: If the connection on the H-bundle is real-analytic, then the map into BH may be chosen to be real-analytic. If the connection is invariant under some compact Lie group G, then the map into BH may be chosen to be equivariant with respect to G. Furthermore, these maps are unique up to a connection preserving homotopy.

§1. The Orthogonal Group

The classifying space for $O(k)$ is the direct limit (as n tends to infinity) of the Grassmanian manifolds $G_k(\mathbb{R}^n) = O(n)/O(k)\times O(n-k)$. The total space of the canonical principal $O(k)$-bundle on $G_k(\mathbb{R}^n)$ is the Stiefel manifold $O(n)/O(n-k)$. The left-invariant (Maurer-Cartan) form $A^{-1}dA$ on $O(n)$ determines a k-by-k submatrix of 1-forms ω which is well-defined on $O(n)/O(n-k)$ and is called the canonical connection (see [2]). This connection is compatible with the direct limit.

*Department of Mathematics, University of California at Berkeley.

This can be described more geometrically in terms of the associated vector bundle $\pi: \gamma^k(\mathbb{R}^n) \to G_k(\mathbb{R}^n)$. We have

$$G_k(\mathbb{R}^n) = \{W \subset \mathbb{R}^n \mid W \text{ is a subspace of dimension } k\}$$
$$\gamma^k(\mathbb{R}^n) = \{(v,W) \mid v \in W \text{ and } W \in G_k(\mathbb{R}^n)\}$$

with $\pi(v,W) = W$. The horizontal curves through (v,W) are given by "rotations of v perpendicular to W," that is

$$t \mapsto (\exp(tB)v, \exp(tB)W)$$

where B belongs to the subspace of $o(n)$ that can be identified with $\mathrm{Hom}(W,W^{\perp})$.

The proof of the Narasimhan-Ramanan theorem is motivated by the following well-known

Proposition: *Given a submanifold M^k of $\mathbb{R}^n$, the connection on M induced from $\mathbb{R}^n$ is the same as the one pulled back from $\gamma^k(\mathbb{R}^n)$ via the Gauss map $M \to G_k(\mathbb{R}^n)$.*

Proof: On a small neighborhood in $\mathbb{R}^n$ take an orthogonal (adapted) moving frame $v_1,\ldots,v_n$ such that, at points in M, $v_1,\ldots,v_k$ are tangent to M. There is a unique matrix of 1-forms $\omega = (\omega_{ij})$ with $dv_j = \sum_{i=1}^n v_i\omega_{ij}$. Let $A = (v_1,\ldots,v_n) \in O(n)$ so, in matrix notation, $dA = A\cdot\omega$ or $\omega = A^{-1}dA$. Thus $(\omega_{ij})_{i,j=1,\ldots,k}$ is just the pull-back of the universal connection with respect to the moving frame. But this agrees with the connection induced from $\mathbb{R}^n$, since on M we have $\nabla v_j = \sum_{i=1}^k v_i\omega_{ij}$.

Theorem. *Let M be a smooth m-dimensional manifold with a k-dimensional vector bundle E. Suppose that E has a (fiber) metric h and a connection compatible with h. Then for $n \geq \frac{1}{2}((k+m)^2 + 7(k+m) + 10)$ there exists*

a smooth map $f: M \to G_k(\mathbb{R}^n)$ *such that* E *and* $f^*\gamma^k(\mathbb{R}^n)$ *are isomorphic as bundles with metrics and connections. If the metric and connection on* E *are real-analytic then the map* f *may be taken to be analytic.*

Proof. Let g be any Riemannian metric on M. (If M is analytic, we may choose g to be analytic). Make the total space of E into a Riemannian manifold by taking the orthogonal direct sum of g and h; i.e. use g on horizontal vectors and h on vertical vectors. By the Nash imbedding theorem E imbeds isometrically in $\mathbb{R}^n$ where, according to Gromov-Rohklin [1], we may take $n = \frac{1}{2}((k+m)^2 + 7(k+m) + 10)$. (This imbedding may be taken to be analytic if E is analytic). Define a map $E \to G_k(\mathbb{R}^n)$ as follows: Given a point in E, take the tangent space (translated to the origin) to the fiber at its image in $\mathbb{R}^n$. Composing this with the zero section gives the desired map.

We must now show that the connection pulled back from the universal connection is the same as the connection that we started with. By the argument of the above proposition, the pull-back connection is the same as the one induced from $\mathbb{R}^n$, and hence the same as the one induced from the Riemannian structure of E. The proof is completed by the following proposition which shows that, as vector bundles with metrics and connections, E is canonically identified with the normal bundle to the zero-section.

Proposition. *Let* $\tilde{\nabla}$ *be the Riemannian connection on* E. *Then for* $X \in TM \subset TE$,

(1) *if* Y *is tangent to* M, *so is* $\tilde{\nabla}_X Y$;

(2) *if* Y *is normal to* M, *so is* $\tilde{\nabla}_X Y$;

(3) *the induced connections on the tangent and normal bundles to* M *agree with the original connections.*

Proof. Let $\{x^i\}$ be a local coordinate system on M and let $\{e_\alpha\}$ be an orthonormal moving frame of local sections to the bundle $E \to M$. We get local coordinates on E by letting $(x^1(x),\dots,x^m(x),y^1,\dots,y^k)$ represent the point $\sum y^\alpha e_\alpha(x) \in E$. Let $\{\Gamma^\alpha{}_{i\beta}\}$ be the connection coefficients of the given connection ∇ on E, so $\nabla_{\partial/\partial x^i} e_\beta = \sum_{\alpha=1}^{k} \Gamma^\alpha{}_{i\beta} e_\alpha$. Note that $\Gamma_{\alpha i\beta} = \Gamma^\alpha{}_{i\beta} = -\Gamma^\beta{}_{i\alpha}$ since the frame $\{e_\alpha\}$ is orthonormal. A vector

$$\sum a^i \frac{\partial}{\partial x^i} + \sum b^\alpha \frac{\partial}{\partial y^\alpha}$$

in TE has horizontal component

$$\sum a^i \frac{\partial}{\partial x^i} + \sum a^i y^\beta \Gamma^\gamma{}_{i\beta} \frac{\partial}{\partial y^\gamma}$$

so the Riemannian metric on E is given by

$$\sum g_{ij} dx^i dx^j + \sum \delta_{\alpha\beta} dy^\alpha dy^\beta + \sum y^\beta \Gamma_{\alpha i\beta} dx^i dy^\alpha \ .$$

Let $\tilde{\Gamma}$ be the Levi-Civita connection for this metric, restricted to M. Then, by direct computation,

$$\begin{aligned}
\tilde{\Gamma}_{ijk} &= \tfrac{1}{2}(\partial g_{ij}/\partial x^k + \partial g_{ik}/\partial x^j - \partial g_{jk}/\partial x^i) \\
\tilde{\Gamma}_{\alpha i\beta} &= \Gamma_{\alpha i\beta} \\
\Gamma_{\alpha ij} &= 0 = \Gamma_{ij\alpha} \ .
\end{aligned}$$

The result follows.

§2. The Narasimhan-Ramanan Theorem

Let H be any compact Lie group, which we may suppose to be imbedded in $O(k)$ for some k. Let $\pi\colon o(k) \to \mathfrak{h}$ be the projection determined by the killing form. Let ω be the 1-form on $O(n)/O(n-k)$ with values in $o(k)$ which was defined in §1. Then $\pi\circ\omega$ is a connection on the principal

H-bundle

$$O(n)/O(n-k) \longrightarrow O(n) / H \times O(n-k) \ .$$

Taking the direct limit as n tends to infinity gives the classifying space BH along with its canonical principal H-bundle and its canonical connection.

Theorem. *Suppose that* $\dim M \leq m$ *and* $n \geq \frac{1}{2}((k+m)^2 + 7(k+m) + 10)$. *Then every principal* H*-bundle with connection over* M *may be obtained as the pull-back of the canonical principal* H*-bundle and canonical connection over* $O(n) / H \times O(n-k)$ *by some map* $f: M \longrightarrow O(n) / H \times O(n-k)$. *If the bundle with connection over* M *is real-analytic, then the map* f *may be taken to be analytic.*

Proof. When $H = O(k)$, this theorem is equivalent to the theorem in §1. To deduce the general case, suppose $P \longrightarrow M$ is a principal H-bundle with connection. Let $\tilde{P} = P \times_H O(k)$ be the corresponding principal $O(k)$-bundle. The induced connection on $\tilde{P}$ is the pull-back of ω under some $O(k)$-map. Thus we have a sequence of connection preserving H-maps

$$P \longrightarrow \tilde{P} \longrightarrow O(n)/O(n-k) \ .$$

The pull-back of ω has values in $\mathfrak{h} \subseteq o(k)$, so we could use $\pi \circ \omega$ instead of ω. Dividing out by H gives the desired map from M to $O(n)/ H \times O(n-k)$. For more details, see [4].

In this theorem the maps from M to $O(n)/ H \times O(n-k)$ are unique in the sense that there is a connection preserving homotopy in $O(2n)/ H \times O(2n-k)$ between any two such maps. Furthermore we have a natural one-to-one correspondence between equivalence classes of principal H-bundles with connections and connection preserving homotopy classes of maps from M to BH. The

proof of these assertions follows from the methods of §4 where a slightly different construction of BH is used.

§3. An Equivariant Version of the Narasimhan-Ramanan Theorem

The following equivariant version of the Nash imbedding theorem was recently proved.

Theorem [3]. *If* M *is a compact Riemannian manifold (possibly with boundary) and* G *is a compact Lie group acting on* M *by isometries, then there is an orthogonal representation* ρ *of* G *on some Euclidean space* $\mathbb{R}^n$ *and an isometric imbedding of* M *into* $\mathbb{R}^n$ *which is equivariant with respect to* ρ.

Let G and H be compact Lie groups, and consider H to be a subgroup of some $O(k)$. A G-manifold is a smooth manifold with a smooth action by G. A G-map is a smooth map of G-manifolds which commutes with the action of G.

Definition. A (G,H)-bundle on a G-manifold M consists of

(1) a principal H-bundle $P \to M$;

(2) a connection on $P \to M$;

(3) an action of G on P such that

  (a) G commutes with H,

  (b) the induced action of G on M agrees with the original one,

  (c) G preserves the connection.

Examples: (1) A G-manifold M^m with a G-invariant metric has a natural $(G,O(m))$-bundle given by the bundle of orthonormal frames.

(2) The canonical bundle over $O(n)/H \times O(n-k)$ is a $(O(n),H)$-bundle.

(3) Any representation $\rho: G \to O(n)$ determines a canonical (G,H)-bundle over $O(n)/H \times O(n-k)$.

Theorem. *For any* (G,H)*-bundle on a compact* G*-manifold* M*, there exists a representation* $\rho: G \to O(n)$ *and a* G*-map* $f: M \to O(n)/H \times O(n-k)$ *such that the* (G,H)*-bundle on* M *is isomorphic to the* (G,H)*-bundle induced by* f *and* ρ*. If the* (G,H)*-bundle on* M *is real-analytic then* f *may be chosen to be analytic.*

Proof. By imbedding the (G,H)-bundle in a $(G,O(k))$-bundle, it will suffice to treat the case where $H = O(k)$. Let $P \to M$ be the given principal $O(k)$-bundle, and give M a G-invariant Riemannian metric. Let $E = P \times_{O(k)} \mathbb{R}^k$ be the associated vector bundle and let E_1 be the unit disc bundle. Defining a Riemannian metric on E as before, it is easily checked that G acts on E_1 by isometries. Then for some representation $\rho: G \to O(n)$ there is an equivariant isometric imbedding of E_1 into $\mathbb{R}^n$. As before define $f: M \to G_k(\mathbb{R}^n)$ by looking at the tangent k-plane to the fiber in $\mathbb{R}^n$. The method of §1 shows that f has the required properties.

§4. A Uniqueness Theorem

Let G be a compact Lie group and let $\mathcal{H}$ be $L^2(\mathbb{N})$, the real Hilbert space of square summable sequences. Let P_k be the set of k-tuples of orthonormal vectors in $\mathcal{H}$, with the obvious action of $O(k)$ and $O(\mathcal{H})$. If H is a closed subgroup of $O(k)$, let BH be P_k/H. With either the strong or the weak topology, BH is the classifying space for H. With the strong topology, BH is a Banach manifold. For any continuous representation of G on $\mathcal{H}$, BH has a canonical (G,H)-bundle.

In particular, if $H = O(k)$ then

$$BO(k) = \{W \subset \mathcal{H} \mid W \text{ is a subspace of dimension } k\}$$

and the tangent bundle is given by

$$T_W BO(k) = L(W, W^{\perp})$$

where $L(W,W^{\perp})$ is the space of linear maps from W to $W^{\perp}$.

Let M be a G-manifold. Two (G,H)-bundles are isomorphic if there exists a bundle isomorphism that preserves the connection and commutes with G. A one-parameter family of G-maps $f_t: M \to BH$ is a connection preserving G-homotopy if for each t_1 and t_2 the (G,H)-bundles induced by f_{t_1} and f_{t_2} are isomorphic. The main theorem of this section is:

Theorem: *The inducing process gives a one-to-one correspondence between isomorphism classes of* (G,H)*-bundles on* M *and connection preserving* G*-homotopy classes of maps from* M *to* BH.

Proof. We first prove the theorem when $H = O(k)$. It is necessary to show that any $(G,O(k))$-bundle on any G-manifold M is induced by a map into $BO(k)$, and that there is a connection preserving G-homotopy between any two such maps. The first assertion follows from the technique of §3 and the theorem that any G-manifold admits an equivariant isometric imbedding into $\mathcal{H}$ (see [3]).

Suppose we have smooth maps $f_0, f_1: M \to BO(k)$ which are equivariant with respect to representations $\rho_0, \rho_1: G \to O(\mathcal{H})$ and an isomorphism of the induced $(G,O(k))$-bundles. We may think of the $(G,O(k))$-bundles as vector bundles with fiber metrics, connections, and G-actions. Having an isomorphism of the induced $(G,O(k))$-bundles means that for each $x \in M$ we have an isometry

$$h(x)\colon f_0(x) \to f_1(x)$$

where h depends smoothly on x. Furthermore h is equivariant, i.e.

$$h(\sigma x)\rho_0(\sigma) = \rho_1(\sigma)h(x) \ , \quad x \in M, \ \sigma \in G \ ,$$

and h maps horizontal curves to horizontal curves. A curve $c(s)$ in the vector bundle induced by f_0 lies over a curve $x(s)$ in M if $c(s) \in f_0(x(s))$ for all s. $c(s)$ is horizontal if

$$\frac{d}{ds}\, c(s) \perp f_0(x(s))$$

for all s. The condition on h then requires that

$$\frac{d}{ds}\, h(x(s))c(s) \perp f_1(x(s)) \ .$$

Define $\theta_n(t) = \frac{\pi}{2}n(n+1)t - \frac{\pi}{2}n$ and define a partial isometry $A(t)\colon \mathcal{H} \to \mathcal{H}$ by

$$A(t)\colon (a_0,a_1,\ldots) = (a_0,\ldots,a_{n-2},a_{n-1}\cos\theta_n(t),a_{n-1}\sin\theta_n(t),$$
$$a_n\cos\theta_n(t),a_n\sin\theta_n(t),\ldots)$$

for $\frac{1}{n+1} \le t \le \frac{1}{n}$ and let $A(0)$ be the identity. $A(t)$ is continuous in t and satisfies

$$A(\tfrac{1}{2})(a_0,a_1,\ldots) = (a_0,0,a_1,0,\ldots)$$
$$A(1)(a_0,a_1,\ldots) = (0,a_0,0,a_1,\ldots)$$

Define a representation μ_t to be the direct sum of the representation $A(t)\circ\rho_0\circ A(t)^{-1}$ on Ran $A(t)$ and the representation on Ran $A(t)^{\perp}$ induced by ρ_1 and the isomorphism Ran $A(t)^{\perp} \cong \mathcal{H}$ given by

$$(-1)^n(0,\ldots,0,-a_{n-1}\sin\theta_n(t),a_{n-1}\cos\theta_n(t),-a_n\sin\theta_n(t),$$
$$a_n\cos\theta_n(t),\ldots) \longmapsto (a_{n-1},a_n,\ldots)$$

for $\frac{1}{n+1} \le t \le \frac{1}{n}$. This satisfies

$$\mu_t \circ A(t) = A(t)\circ\rho_0$$

and

$$\mu_1 \circ A(\tfrac{1}{2}) = A(\tfrac{1}{2})\circ\rho_1 \ .$$

Each $A(t)$ is a partial isometry, so $A(t)f_0$ is a connection preserving G-homotopy, being equivariant with respect to μ_t. Similarly, f_1 is connection preserving G-homotopic to $A(\frac{1}{2})f_1$, which is equivariant with respect to μ_1. Let

$$g_t(x) = [\cos t\, A(1) + \sin t\, A(\tfrac{1}{2})h(x)]f_0(x) \ .$$

This is equivariant with respect to μ_1, and is a connection preserving G-homotopy from $A(1)f_0$ to $A(\frac{1}{2})f_1$. This completes the proof when $H = O(k)$.

Now suppose that H is an arbitrary closed subgroup of $O(k)$. A given (G,H)-bundle on M may be enlarged to a $(G,O(k))$-bundle, so it is induced by some G-map of M into BH as before. Furthermore, it is not hard to show that any connection preserving G-map from a (G,H)-bundle to BH must factor through this enlargement. The theorem follows easily.

References

1. Gromov, M. L., Rokhlin, V. A.: Embeddings and immersions in riemannian geometry. Russian Math. Surveys 25, 1-57 (1970).

2. Kobayashi, S., Nomizu, K.: Foundations of Differential Geometry, Volume II. New York: Interscience, 1969.

3. Moore, J. D., Schlafly, R.: On equivariant isometric embeddings. (To appear)

4. Narasimhan, M. S., Ramanan, S.: Existence of universal connections. Amer. J. Math. 83, 563-572 (1961).

5. Steenrod, N.: The Topology of Fibre Bundles. Princeton University Press, 1951.

Roger Schlafly
Department of Mathematics
University of California
Berkeley, CA 94720
U.S.A.

Chapter II

Universal Connections: The Local Problem

by Roger Schlafly

§1. Introduction

In this paper we consider connections on a principal G-bundle over a manifold M of dimension m, where G is O(n), U(n), or Sp(n). The universal examples are the bundles

$$O(N)/O(N-n) \longrightarrow O(N)/O(n)\times O(N-n)$$
$$U(N)/U(N-n) \longrightarrow U(N)/U(n)\times U(N-n)$$
$$SP(N)/Sp(N-n) \longrightarrow SP(N)/Sp(n)\times Sp(N-n)$$

with their canonical connections. (See section 2 for the definitions.) These are the Stiefel and Grassman manifolds for $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{H}$. It is a theorem of Narasimhan and Ramanan that the canonical connection is universal in the sense that any connection on a principal G-bundle over M^m is induced by a map into the appropriate Grassman manifold N is sufficiently large. According to [2], [4], and section 8, it suffices to take

$$N \geq 2n(m+1)(2mn^2+1) \quad \text{or} \quad \tfrac{1}{2}[(n+m)^2+7(n+m)+10]$$

for O(n),

$$N \geq n(m+1)(2mn^2+1)$$

for U(n), and

$$N \geq n(m+1)(4mn^2+2mn+1)$$

for Sp(n).

These inequalities are not sharp. The situation is analogous to the problem of finding isometric imbeddings of riemannian manifold into euclidean space. In that case, global results are available, but they are not sharp with respect to dimension. The only sharp results date back to L. Schläfli [3] who found the least dimensional euclidean space for the local isometric imbeddability of a real-analytic riemannian manifold. This was made more rigorous by M. Janet, C. Burstin, and E. Cartan. (See [5] for an account of this theorem.)

The principal result of this paper is to show how E. Cartan's theory of differential systems can be used to get local existence theorems for connection preserving maps into the appropriate Grassmanian. These results are sharp with respect to dimension.

I would like to thank my advisor, I. M. Singer, for suggesting this problem to me. I am also grateful to S. S. Chern for explaining differential systems. Finally, I am indebted to Uncle Ludwig for inspiration.

§2. Definition of the Universal Connection

Let F be $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{H}$. The Stiefel manifold S is the set of N-by-n matrices P over F such that $P^*P = 1$. For $P \in S$, let

$$W_p = \text{im } PP^*$$

so PP^* is the projection from F^N onto W_p. The map $P \mapsto W_p$ gives a fibration of S over the Grassman manifold

$$\{W \subseteq F^N : W \text{ is a subspace of dimension } n\} .$$

It is a principal bundle with the obvious action of $U(n,F)$ (i.e., $O(n)$,

$U(n)$, or $Sp(n)$). It has a canonical connection defined as follows: the horizontal space at P is $\hom_F(W_p, W_p^{\perp})$ where T_pS is identified with the subspace of N-by-N matrices A such that $A + A^* = 0$.

The connection one-form is given by

$$\psi(A) = P^*AP \ , \quad A \in T_pS \ .$$

The curvature is

$$\Phi(A,B) = -P^*[A,B]P + [P^*AP, P^*BP] \ , \quad A, B \in T_pS \ .$$

Let o be the point

$$\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix}$$

in S. The horizontal vectors at o can be identified with $(N-n)$-tuples of vectors in F^n as follows: $(a_1, \ldots, a_{N-n}) \in F^n \times \cdots \times F^n$ is identified with

$$\begin{bmatrix} 0 & a_1 & \cdots & a_{N-n} \\ -a_1^* & & & \\ \vdots & & & \\ -a_{N-n}^* & & & \end{bmatrix} .$$

The curvature form is

$$\Phi((a_1, \ldots, a_{N-n}), (b_1, \ldots, b_{N-n})) = a_1 \wedge b_1 + \cdots + a_{N-n} \wedge b_{N-n}$$

where

$$a \wedge b = ab^* - b^*a \ , \quad a, b \in F^n$$

so $a \wedge b$ is an n-by-n matrix.

§3. Cartan's Method

Suppose we have a real-analytic connection on a principal U(n,F)-budnle over $\mathbb{R}^m$. With respect to a moving frame, the connection is given by a skew-hermitian matrix $\omega = (\omega_{ij})$ of one-forms on $\mathbb{R}^m$. The curvature is

$$\Omega = (\Omega_{ij}) = d\omega + \omega \wedge \omega \ .$$

We want to find a map

$$f: \mathbb{R}^m \longrightarrow U(N,F)/U(N-n,F)$$

such that in a neighborhood of $o \in \mathbb{R}^m$,

$$\omega = f^*\psi \ .$$

The graph of f will be a submanifold of $\mathbb{R}^m \times U(N,F)/U(N-n,F)$ containing $(0,o)$, having dimension m, and transverse to $U(N,F)/U(N-n,F)$. The one-forms

$$\pi_1^*\omega_{ij} - \pi_2^*\omega_{ij} \ , \quad 1 \leq i,j \leq n$$

vanish on this submanifold, where π_1 and π_2 are the obvious projections. Furthermore, our given connection may be obtained from the universal connection if and only if we can find a submanifold with these properties.

Let I be the ideal of differential forms generated by

$$\pi_1^*\omega_{ij} - \pi_2^*\psi_{ij} \ , \quad 1 \leq i,j \leq n$$

and

$$\pi_1^*\Omega_{ij} - \pi_2^*\Phi_{ij} , \quad 1 \le i,j \le n .$$

Lemma 3.1. (a) I is closed under the exterior derivative.

(b) If the forms $\pi_1^*\omega_{ij} - \pi_2^*\psi_{ij}$ vanish on a submanifold, then all of I does.

Proof. (a) We have, by definition,

$$\Omega_{ij} = d\omega_{ij} + \sum_k \omega_{ik} \wedge \omega_{kj}$$
$$\Phi_{ij} = d\psi_{ij} + \sum_k \psi_{ik} \wedge \psi_{kj} ,$$

so

$$\begin{aligned} d(\pi_1^*\omega_{ij} - \pi_2^*\psi_{ij}) &= \pi_1^* d\omega_{ij} - \pi_2^* d\psi_{ij} \\ &= \pi_1^*(\Omega_{ij} - \sum_k \omega_{ik} \wedge \omega_{kj}) - \pi_2^*(\Phi_{ij} - \sum_k \psi_{ik} \wedge \psi_{kj}) \\ &= \pi_1^*\Omega_{ij} - \pi_2^*\Phi_{ij} \\ &\quad - \sum_k \pi_1^*\omega_{ik} \wedge (\pi_1^*\omega_{kj} - \pi_2^*\psi_{kj}) - \sum_k (\pi_1^*\omega_{ik} - \pi_2^*\psi_{ik}) \wedge \pi_2^*\psi_{kj} . \end{aligned}$$

Also, the Bianchi identities are

$$d\Omega_{ij} + \sum_k \omega_{ik} \wedge \Omega_{kj} - \Omega_{ik} \wedge \omega_{kj} = 0$$
$$d\Phi_{ij} + \sum_k \psi_{ik} \wedge \Phi_{kj} - \Phi_{ik} \wedge \psi_{kj} = 0 ,$$

so

$$\begin{aligned} d(\pi_1^*\Omega_{ij} - \pi_2^*\Phi_{ij}) &= \pi_1^* d\Omega_{ij} - \pi_2^* d\Phi_{ij} \\ &= \pi_1^* \sum_k \Omega_{ik} \wedge \omega_{kj} - \omega_{ik} \wedge \Omega_{kj} \\ &\quad - \pi_2^* \sum_k \Phi_{ik} \wedge \psi_{kj} - \psi_{ik} \wedge \Phi_{kj} \\ &= \sum_k (\pi_1^*\Omega_{ik} - \pi_2^*\Phi_{ik}) \wedge \pi_1^*\omega_{kj} + \sum_k \pi_2^*\Phi_{ik} \wedge (\pi_1^*\omega_{kj} - \pi_2^*\psi_{kj}) \\ &\quad - \sum_k \pi_1^*\omega_{ik} \wedge (\pi_1^*\Omega_{kj} - \pi_2^*\Phi_{kj}) - \sum_k (\pi_1^*\omega_{ik} - \pi_2^*\psi_{ik}) \wedge \pi_2^*\Phi_{kj} . \end{aligned}$$

It follows easily that $dI \subset I$.

(b) Let f be the inclusion map of a submanifold. If

$$f^*(\pi_1^*\omega_{ij} - \pi_2^*\psi_{ij}) = 0 , \quad 1 \leq i,j \leq n ,$$

then

$$\begin{aligned} f^*(\pi_1^*\Omega_{ij} - \pi_2^*\Omega_{ij}) &= f^*d(\pi_1^*\omega_{ij} - \pi_2^*\psi_{ij}) \\ &= d[f^*(\pi_1^*\omega_{ij} - \pi_2^*\psi_{ij})] \\ &= 0 . \end{aligned}$$ //

Thus I is a differential system and our problem is equivalent to finding an integral submanifold for I. The Cartan-Kähler theorem gives a sufficient condition for such a submanifold to exist (see [1] or [5] for modern treatment). For the rest of this paper we assume that all connections are real-analytic.

Let $e_1,\ldots,e_m$ be the standard basis for $\mathbb{R}^m$. It is not hard to show that for N sufficiently large, every

$$A \in \Lambda^2(\mathbb{R}^m)\otimes u(n,F)$$

is of the form

$$A = \frac{1}{2} \sum_{i,j=1}^{m} \sum_{k=1}^{N-n} (e_i \wedge e_j)\otimes(a_i^k \wedge a_j^k) \tag{3.2}$$

for some $a_i^k \in F^n$. Let N_{mn} be the least such integer N.

We now discuss the problem of finding the least N such that every connection on a $U(n,F)$-bundle over $\mathbb{R}^m$ is obtained from a map

$$f: \mathbb{R}^m \longrightarrow U(N,F)/U(N-n,F) .$$

A necessary condition is that $N \geq N_{mn}$ because if A is the value of $f^*\Phi$ at some point, then A satisfies equation (3.2) where

$(a_i^1,\ldots,a_i^{N-n})$ corresponds to the horizontal part of $f_*(e_i)$. Conversely, at least in the generic case, the Cartan-Kähler theorem may be used to show that we may actually take $N = N_{mn}$. In the following four sections, this procedure is worked out for specific examples.

§4. O(n)-Bundles over $\mathbb{R}^2$

Theorem 4.1. Every O(n)-bundle over $\mathbb{R}^2$ is locally obtained from a map

$$f: \mathbb{R}^2 \longrightarrow O(N)/O(N-n)$$

if and only if $N \geq \frac{3n-1}{2}$.

Proof. Suppose n is 2k or 2k+1, and $N = n+k$ so $N \geq \frac{3n-1}{2}$. We are going to apply the Cartan-Kähler theorem, so the reader is referred to [1] or [5] for the relevant definitions.

Each point is a regular integral element, as its polar space always has dimension two.

Choose $a_1,\ldots,a_k$, $b_1,\ldots,b_k$ in $\mathbb{R}^n$ such that

$$\Omega(e_1,e_2) = a_1\wedge b_1 + \cdots + a_k\wedge b_k \ .$$

That we can do this follows from the normal form theorem for skew-symmetric bilinear forms. Furthermore, we can arrange $a_1,\ldots,a_k$ to be linearly independent.

Let Y_1, $Y_2 \in T_0O(n+k)/O(k)$ be the vectors with vertical parts $\omega(e_1)$, $\omega(e_2)$ (resp.) and horizontal parts $(a_1,\ldots,a_k)$, $(b_1,\ldots,b_k)$ (resp.). The polar space for (e_1,Y_1) is

$$\varepsilon(e_1,Y_1) = \{(X,Y):\ \omega(X) = \psi(Y),\ \Omega(e_1,X) = \Phi(Y_1,Y)\} \ .$$

It is not hard to see that this has constant dimension $k+2+k(k-1)/2$, so (e_1,Y_1) is a regular integral element.

Thus we can now apply the Cartan-Kähler theorem to get an integral submanifold tangent to (e_1,Y_1) and (e_2,Y_2). The existence of f follows. //

§5. SU(2) and U(2) Bundles over $\mathbb{R}^2$

Lemma 5.1. Suppose $M \in u(2)$ has eigenvalues $\lambda_1 i$, $\lambda_2 i$ with $\lambda_1 \leq \lambda_2$. Then there exist $a, b \in \mathbb{C}^2$ such that $a \neq 0$ and

$$M = a \wedge b \tag{5.2}$$

if and only if $\lambda_1 \leq 0 \leq \lambda_2$.

Proof. Suppose that (5.2) holds for some $a, b \in \mathbb{C}^2$. We can assume that $|a| = 1$. Choose $u \in U(2)$ such that

$$ua = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Define $\alpha, \beta, \gamma, \delta \in \mathbb{R}$ by

$$\begin{pmatrix} \alpha+i\beta \\ \gamma+i\delta \end{pmatrix} = ub .$$

Then

$$\begin{aligned} uMu^* &= u(a \wedge b)u^* \\ &= ua \wedge ub \\ &= \begin{pmatrix} 0 \\ 1 \end{pmatrix} \wedge \begin{pmatrix} \alpha+i\beta \\ \delta+i\delta \end{pmatrix} \\ &= \begin{pmatrix} 0 \\ 1 \end{pmatrix} \wedge \begin{pmatrix} \alpha+i\beta \\ i\delta \end{pmatrix} \\ &= \begin{pmatrix} 0 & -\alpha-i\beta \\ \alpha-i\beta & -2i\delta \end{pmatrix} \end{aligned}$$

has eigenvalues $\lambda_1 i$ and $\lambda_2 i$, so

$$\begin{aligned} \lambda_1 &= -\delta - \sqrt{\alpha^2+\beta^2+\delta^2} \\ \lambda_2 &= -\delta + \sqrt{\alpha^2+\beta^2+\delta^2} \ . \end{aligned} \tag{5.3}$$

It follows that $\lambda_1 \leq 0 \leq \lambda_2$.

Conversely, if $\lambda_1 \leq 0 \leq \lambda_2$ then we can find $\alpha, \beta, \delta \in \mathbb{R}$ such that (5.3) holds. Therefore

$$M = u\left[\begin{pmatrix} 0 \\ 1 \end{pmatrix} \wedge \begin{pmatrix} \alpha+i\beta \\ i\beta \end{pmatrix}\right]u^*$$

for some $u \in U(2)$, and hence

$$M = u\begin{pmatrix} 0 \\ 1 \end{pmatrix} \wedge u\begin{pmatrix} \alpha+i\beta \\ i\delta \end{pmatrix} \ . \qquad //$$

Now suppose we have a connection ω on a U(2)-bundle over $\mathbb{R}^2$, with curvature Ω. Let $\lambda_1 i$, $\lambda_2 i$ $(\lambda_1 \leq \lambda_2)$ be the eigenvalues of $\Omega(e_1,e_2)$.

Theorem 5.4. On a neighborhood V of $0 \in \mathbb{R}^2$ the connection is obtained from a map

$$f: V \longrightarrow U(4)/U(2)\times U(2) \ .$$

In order for there to exist a neighborhood V of $o \in \mathbb{R}^2$ such that the given connection is obtained from a map

$$g: V \longrightarrow U(3)/U(2)\times U(1)$$

it is necessary and sufficient that there exist a neighborhood of $o \in \mathbb{R}^2$ such that $\lambda_1 \leq 0 \leq \lambda_2$.

Proof. Suppose

$$u^*\Omega(e_1,e_2)u = \begin{pmatrix} \lambda_1 i & 0 \\ 0 & \lambda_2 i \end{pmatrix} .$$

If we let

$$a = u\begin{pmatrix} 1 \\ 0 \end{pmatrix} , \qquad c = u\begin{pmatrix} 0 \\ 1 \end{pmatrix} ,$$
$$b = -\tfrac{1}{2}\lambda_1 ia , \qquad d = -\tfrac{1}{2}\lambda_2 ic ,$$

then a and c are linearly independent and

$$\Omega(e_1,e_2) = a\wedge b + c\wedge d .$$

Let Y_1, $Y_2 \in T_oU(4)/U(2)$ be the vectors with vertical parts $\omega(e_1)$, $\omega(e_2)$ (resp.) and horizontal parts (a,c), (b,d) (resp.). Then, as in the proof of Theorem 4.1,

$$(0,o) < (e_1,Y_1) < \{(e_1,Y_1),(e_2,Y_2)\}$$

is a chain of ordinary integral elements satisfying the hypothesis of the Cartan-Kähler theorem. Hence an appropriate integral submanifold exists.

The second part of Theorem 5.4 follows similarly, given Lemma 5.1. //

Corollary 5.5. Every connection on a SU(2)-bundle over $\mathbb{R}^2$ comes locally from a map

$$g: \mathbb{R}^2 \longrightarrow U(3)/U(2)\times U(1) .$$

Connections on U(1)-bundles over $\mathbb{R}^2$ come locally from a map

$$h: \mathbb{R}^2 \longrightarrow U(2)/U(1)\times U(1) .$$

§6. O(2) and O(3) Bundles over $\mathbb{R}^3$

Let ω be a connection on an O(2)-bundle over $\mathbb{R}^3$, with curvature Ω.

Theorem 6.1. Suppose $\Omega_0 \neq 0$. Then there exists a neighborhood V of 0 such that ω is induced by a map

$$f: V \longrightarrow O(3)/O(2)\times O(1) .$$

Proof. Let

$$\Omega(e_i,e_j) = \begin{pmatrix} 0 & \alpha_{ij} \\ -\alpha_{ij} & 0 \end{pmatrix}$$

where we have chosen our basis e_1, e_2, e_3 for $\mathbb{R}^3$ so that $\alpha_{12} \neq 0$. Let

$$a_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$a_2 = \begin{pmatrix} 0 \\ \alpha_{12} \end{pmatrix}$$

$$a_3 = \begin{pmatrix} -\alpha_{23}/\alpha_{12} \\ \alpha_{13} \end{pmatrix} .$$

Then we have

$$\Omega(e_i,e_j) = a_i \wedge a_j .$$

Let $Y_i \in T_0O(3)/O(1)$ have vertical part $\omega(e_i)$ and horizontal part a_i. Since a_1 and a_2 are linearly independent, we have a chain of ordinary integral elements:

$$(0,0) < (e_1,Y_1) < \{(e_1,Y_1),(e_2,Y_2)\}$$
$$< \{(e_1,Y_1),(e_2,Y_2),(e_3,Y_3)\} .$$

Thus the Cartan-Kähler theorem applies. //

Define $T: \mathbb{R}^3 \longrightarrow o(3)$ by

$$T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 & z & -y \\ -z & 0 & x \\ y & -x & 0 \end{pmatrix} ,$$

so

$$a \wedge b = T(a \times b) \quad , \quad a, b \in \mathbb{R}^3$$

where $a \times b$ is the usual cross product on $\mathbb{R}^3$.

Lemma 6.2. Suppose $b_1, b_2, b_3 \in \mathbb{R}^3$ are linearly independent. Then there exist vectors a_1, a_2, a_3 in $\mathbb{R}^3$ such that

$$\left.\begin{aligned} a_1 \times a_2 &= b_3 \\ a_2 \times a_3 &= b_1 \\ a_3 \times a_1 &= b_2 \end{aligned}\right\} \tag{6.3}$$

if and only if $\det(b_1, b_2, b_3) > 0$. In this case, a_1, a_2, a_3 must be linearly independent.

Proof. Suppose (6.3) holds. Let $u_1, u_2, u_3 \in \mathbb{R}^3$ be the unique (up to sign) unit vectors such that u_1 is perpendicular to b_2 and b_3, u_2 is perpendicular to b_3 and b_1, and u_3 is perpendicular to b_1 and b_2. Then there exist unique real numbers $\alpha_1, \alpha_2, \alpha_3$, $\beta_1, \beta_2, \beta_3$ so that

$$\begin{aligned} a_1 &= \alpha_1 u_1 & \qquad b_1 &= \beta_1\, u_2 \times u_3 \\ a_2 &= \alpha_2 u_2 & \qquad b_2 &= \beta_2\, u_3 \times u_1 \\ a_3 &= \alpha_3 u_3 & \qquad b_3 &= \beta_3\, u_1 \times u_2 \, . \end{aligned}$$

Equation (6.3) now reduces to

$$\alpha_1\alpha_2 = \beta_3$$
$$\alpha_2\alpha_3 = \beta_1$$
$$\alpha_3\alpha_1 = \beta_2 .$$

These equations imply

$$\beta_2\beta_3/\beta_1 = \alpha_1^2$$
$$\beta_3\beta_1/\beta_2 = \alpha_2^2$$
$$\beta_1\beta_2/\beta_3 = \alpha_3^2 ,$$

which can be solved if and only if $\beta_1\beta_2\beta_3 > 0$. But this holds if and only if $\det(b_1,b_2,b_3) > 0$, since

$$\det(b_1,b_2,b_3) = \beta_1\beta_2\beta_3[\det(u_1,u_2,u_3)]^2 . \qquad //$$

Theorem 6.4. Suppose that ω is a connection on an $O(3)$-bundle over $\mathbb{R}^3$, with curvature Ω, and that

$$\gamma = \det(T^{-1}\Omega(e_2,e_3),T^{-1}\Omega(e_3,e_1),T^{-1}\Omega(e_1,e_2)) \neq 0 .$$

Then on a neighborhood V of 0 in $\mathbb{R}^3$, ω is induced from a map

$$f: V \longrightarrow O(5)/O(2) .$$

In order for there to exist a neighborhood W of 0 in $\mathbb{R}^3$ such that ω is induced by a map

$$g: W \longrightarrow O(4)/O(1) ,$$

it is necessary and sufficient that $\gamma > 0$.

Proof. To find g, it is necessary to find vectors a_1, a_2, a_3 in $\mathbb{R}^3$ such that

$$\Omega(e_i,e_j) = a_i \wedge a_j , \quad 1 \leq i,j \leq 3$$

or equivalently,

$$T^{-1}\Omega(e_i,e_j) = a_i \times a_j \ , \quad 1 \leq i,j \leq 3 \ .$$

By Lemma 6.2, this requires that γ be positive. If γ is positive, then a_1, a_2, a_3 exist, so we can apply Cartan-Kähler as before. The linear independence of a_1, a_2, a_3 guarantees that the resulting integral element be ordinary.

If γ is negative, we need to write the curvature at 0 as

$$\Omega(e_i,e_j) = a_i \wedge a_j + b_i \wedge b_j \ .$$

Let

$$\begin{aligned} c_1 &= T^{-1}\Omega(e_2,e_3) \\ c_2 &= T^{-1}\Omega(e_3,e_1) \\ c_3 &= T^{-1}\Omega(e_1,e_2) \ , \end{aligned}$$

so we want to solve

$$\left.\begin{aligned} c_1 &= a_2 \times a_3 + b_2 \times b_3 \\ c_2 &= a_3 \times a_1 + b_3 \times b_1 \\ c_3 &= a_1 \times a_2 + b_1 \times b_2 \end{aligned}\right\} \tag{6.5}$$

given that $\det(c_1,c_2,c_3) < 0$. Let $b_3 = 0$ and let a_3 be the unique (up to sign) unit vector perpendicular to c_1 and c_2. Let a_1, a_2 be vectors satisfying

$$\begin{aligned} c_1 &= a_2 \times a_3 \\ c_2 &= a_3 \times a_1 \ . \end{aligned}$$

By Lemma 6.2, $c_3 \neq a_1 \times a_2$, so choose b_1 and b_2 with

$$b_1 \times b_2 = c_3 - a_1 \times a_2 \ . \tag{6.6}$$

Thus we can solve (6.5).

I claim that we can arrange that

(i) a_1, b_1, b_2 are linearly independent;

(ii) if $a_2 = ra_1 + sb_1 + tb_2$ then $s + rt \neq 0$.

There is only one line perpendicular to c_2 and c_3, and there exists vector v along that line such that $c_2 = a_3 \times v$. Redefine a_1 as $v + a_3$, so $c_2 = a_3 \times a_1$. Since $a_1 \neq v$, a_1 is not perpendicular to c_3. From (6.6) and the fact that a_1 is perpendicular to $a_1 \times a_2$, it follows that a_1 is not perpendicular to $b_1 \times b_2$. We know that b_1 and b_2 are linearly independent because $b_1 \times b_2 \neq 0$. If a_1 were a linear combination of b_1 and b_2, then a_1 would be perpendicular to $b_1 \times b_2$, a contradiction. Part (i) of the claim follows.

Similarly, we can arrange to have a_2, b_1, b_2 linearly independent. Suppose that

$$a_2 = ra_1 + sb_1 + tb_2 .$$

Then r, s, and t are unique and $r \neq 0$. Part (ii) of the claim follows if $s + rt \neq 0$.

Otherwise, assume that $s + rt = 0$. Replace b_1 with $b_1' = b_1 + \epsilon b_2$ for sufficiently small $\epsilon > 0$. Then (6.5) is still satisfied, and so is part (i) of the claim. Then

$$a_2 = ra_1 + sb_1' + (t-\epsilon)b_2$$

is the unique relation of its type. This completes the proof of part (ii) of the claim since

$$s + r(t-\epsilon) = -r\epsilon \neq 0 .$$

Define $Y_i \in T_0O(5)/O(2)$ to have vertical part $\omega(e_i)$ and horizontal part (a_i,b_i). The existence of f now follows once we show that

$$\{(e_1,Y_1),(e_2,Y_2),(e_3,Y_3)\}$$

is an ordinary integral element. It is, by definition, an integral element.

As before, the point $(0,o)$ is regular. To show that (e_1,Y_1) is regular, we must show that

$$\{(d,d')\in\mathbb{R}^6\colon\ a_1\times d + b_1\times d' = 0\}$$

has the least dimension possible, namely three. This happens precisely when a_1 and b_1 are linearly independent. But we know this to be the case by part (i) of the claim.

Finally, we must show that

$$\{(e_1,Y_1),(e_2,Y_2)\}$$

is regular. This means that if

$$a_1 \times d + b_1 \times d' = 0 \qquad (6.7a)$$

$$a_2 \times d + b_2 \times d' = 0 \qquad (6.7b)$$

for some $d, d' \in \mathbb{R}^3$, then $d = d' = 0$.

Suppose that (d,d') is a non-trivial solution to (6.7). If $d' = 0$, then d would be a multiple of a_1 and of a_2, implying that a_1 and a_2 are linearly dependent. But then c_1 and c_2 would be linearly dependent, a contradiction. Thus $d' \neq 0$. Similarly, $d \neq 0$.

It follows from part (i) of the claim that there exists a unique line ℓ through the (a_1,b_1)-plane and the (a_2,b_2)-plane. The general solution to (6.7a) is

$$\begin{aligned} d &= \alpha_1 a_1 + \gamma_1 b_1 \\ d' &= \beta_1 b_1 + \gamma_1 a_1 \end{aligned} \tag{6.8}$$

and the general solution to (6.7b) is:

$$\begin{aligned} d &= \alpha_2 a_2 + \gamma_2 b_2 \\ d' &= \beta_2 b_2 + \gamma_2 a_2 \ . \end{aligned} \tag{6.9}$$

Hence d and d' are along ℓ, so for some nonzero δ, $d' = \delta d$. From (6.7),

$$\begin{aligned} (a_1 + \delta b_1) \times d &= 0 \\ (a_2 + \delta b_2) \times d &= 0 \ . \end{aligned}$$

Hence for some $\mu \in \mathbb{R}$,

$$a_2 + \delta b_2 = \mu(a_1 + \delta b_1) \ .$$

This gives

$$a_2 = \mu a_1 + \mu\delta b_1 - \delta b_2 \ ,$$

which contradicts part (ii) of the claim. Thus $\{(e_1,Y_1),(e_2,Y_2)\}$ is a regular integral element and the hypotheses of the Cartan-Kähler theorem are satisfied. //

§7. O(2)-Bundles over $\mathbb{R}^4$

Let Φ_0 be the 2-form on $O(3)/O(1)$ such that $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\Phi_0$ is the curvature of the canonical connection on the O(2)-bundle

$$O(3)/O(1) \longrightarrow O(3)/O(2)\times O(1) \ .$$

Since $\dim O(3)/O(1) = 3$, $\Phi_0 \wedge \Phi_0 = 0$. Hence if $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\Omega_0$ is the curvature of the connection on the $O(2)$-bundle induced by a map

$$f: \mathbb{R}^4 \longrightarrow O(3)/O(2) \times O(1)$$

then $\Omega_0 \wedge \Omega_0 = 0$.

Theorem 7.1. Suppose we are given a connection ω on an $O(2)$-bundle over $\mathbb{R}^4$ with curvature $\Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\Omega_0$. Assume that $\Omega_0 \neq 0$. Then locally the connection is induced by a map

$$f: \mathbb{R}^4 \longrightarrow O(3)/O(2) \times O(1)$$

if and only if $\Omega_0 \wedge \Omega_0 = 0$.

Proof. Necessity of the condition $\Omega_0 \wedge \Omega_0 = 0$ has already been shown. We now assume that $\Omega_0 \wedge \Omega_0 = 0$. Choose coordinates with $\Omega_0(e_1,e_2) \neq 0$. Let $\alpha_{ij} = \Omega_0(e_i,e_j)$ and

$$a_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad a_2 = \begin{pmatrix} 0 \\ \alpha_{12} \end{pmatrix}, \quad a_3 = \begin{pmatrix} -\alpha_{23}/\alpha_{12} \\ \alpha_{13} \end{pmatrix}$$

$$a_4 = \begin{pmatrix} -\alpha_{24}/\alpha_{12} \\ \alpha_{14} \end{pmatrix}.$$

Since

$$0 = \Omega_0 \wedge \Omega_0 = (\alpha_{12}\alpha_{34} + \alpha_{23}\alpha_{14} - \alpha_{13}\alpha_{24})\mathrm{dvol},$$

it follows that

$$\Omega(e_i,e_j) = a_i \wedge a_j .$$

Let $Y_i \in T\, O(3)/O(1)$ have vertical part $\omega(e_i)$ and horizontal part a_i. Using $\alpha_{12} \neq 0$, it is easy to show that

$$\{(e_1,Y_1),(e_2,Y_2),(e_3,Y_3),(e_4,Y_4)\}$$

is an ordinary integral element, so we can apply the Cartan-Kähler theorem to complete the proof. //

Theorem 7.2. If $\Omega_0 \wedge \Omega_0 \neq 0$ then locally the connection is induced by a map

$$f: \mathbb{R}^4 \to O(4)/O(2)\times O(2) .$$

Proof. Let

$$a_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad a_2 = \begin{pmatrix} 0 \\ \alpha_{12} \end{pmatrix}, \quad a_3 = \begin{pmatrix} -\alpha_{23}/\alpha_{12} \\ \alpha_{13} \end{pmatrix}, \quad a_4 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$b_1 = \begin{pmatrix} \alpha_{14} \\ 0 \end{pmatrix}, \quad b_2 = \begin{pmatrix} \alpha_{24} \\ 0 \end{pmatrix}, \quad b_3 = \begin{pmatrix} \alpha_{23} \\ 0 \end{pmatrix}, \quad b_4 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} .$$

Then

$$\Omega(e_i,e_j) = a_i \wedge a_j + b_i \wedge b_j .$$

Let Y_i have vertical part $\omega(e_i)$ and horizontal part (a_i,b_i). We can thus apply the Cartan-Kähler theorem to the integral element

$$\{(e_1,Y_1),(e_2,Y_2),(e_3,Y_3),(e_4,Y_4)\}$$

once we show it is ordinary. The least trivial step is to show that the matrix

$$\begin{pmatrix} 0 & 1 & 0 & \alpha_{14} \\ -\alpha_{12} & 0 & 0 & \alpha_{24} \\ -\alpha_{13} & -\alpha_{23}/\alpha_{12} & 0 & \alpha_{34} \end{pmatrix}$$

has maximal rank. But this is equivalent to

$$\alpha_{12}\alpha_{34} + \alpha_{14}\alpha_{23} - \alpha_{13}\alpha_{24} \neq 0$$

which is equivalent to the hypothesis

$$\Omega_0 \wedge \Omega_0 \neq 0 \ . \qquad //$$

Chapter III

Universal Connections: Further Remarks

§8. The Quaternionic Grassmanian

We only discuss global results. In [2], Narasimhan and Ramanan show that the canonical connection on the canonical bundle over the real and complex Grassman manifolds is universal. The following theorem shows that their argument extends to the quaternionic case.

Theorem 8.1. If

$$N \geq n(m+1)(4mn^2 + 2mn + 1)$$

then every connection on a principal Sp(n)-bundle over M^m is induced by some map

$$f: M^m \longrightarrow Sp(N)/Sp(n) \times Sp(N-n) .$$

Proof. As in [2], a device using partitions of unity reduces the theorem to the following lemma. //

Lemma 8.2. Let α be a one-form on $B^m = \{x \in \mathbb{R}^m: |x| \leq 1\}$ with values in sp(n). Then there exist n-by-n quaternionic matrix valued functions $\phi_1, \ldots, \phi_{4mn^2+2mn+1}$ on B^m such that

$$\sum_{\ell} \phi_\ell^* \phi_\ell = I$$

and

$$\sum_{\ell} \phi_\ell^* d\phi_\ell = \alpha .$$

Proof. Let $\{f_r\}$ and $\{\tilde{f}_r\}$ be sets of complex n-by-n matrices such that:

(a) Each f_r and $\tilde{f}_r$ is positive definite.

(b) Each f_r and $\tilde{f}_r$ has norm one under the usual operator norm.

(c) $\{f_r\}$ is a basis over $\mathbb{R}$ for the real self-adjoint matrices.

(d) $\{f_r\}$ together with $\{\tilde{f}_r\}$ form a basis over $\mathbb{R}$ for the complex n-by-n self-adjoint matrices.

Let g_r (resp. $\tilde{g}_r$) be the unique positive square root of f_r (resp. $\tilde{f}_r$). We imbed $\mathbb{C}$ into $\mathbb{H}$ in the usual way, i.e. $\mathbb{C} = \mathbb{R} + i\mathbb{R}$ and $\mathbb{H} = \mathbb{R} + i\mathbb{R} + j\mathbb{R} + k\mathbb{R}$. α can be written in the form

$$\alpha = i \sum a_{rs} f_r dx_s + i \sum \tilde{a}_{rs} \tilde{f}_r dx_s + j \sum b_{rs} f_r dx_s + k \sum c_{rs} f_r dx_s$$

where the functions a_{rs}, $\tilde{a}_{rs}$, b_{rs}, c_{rs} are real. Let A be a constant larger than the absolute values of these functions. Let T be the square root of $Amn(2n+1)$.

One can check that

$$I - \frac{3}{n(2n+1)} \sum f_r - \frac{1}{n(2n+1)} \sum \tilde{f}_r$$

is a nonnegative self-adjoint matrix over $\mathbb{C}$, so we can let h be its nonnegative square root. Let $\{\phi_\ell\}$ be the functions:

$$\frac{1}{T}\sqrt{\frac{A + a_{rs}}{2}}\, e^{iTx_s} g_r\,, \quad \frac{1}{T}\sqrt{\frac{A - a_{rs}}{2}}\, e^{-iTx_s} g_r\,,$$

$$\frac{1}{T}\sqrt{\frac{A + \tilde{a}_{rs}}{2}}\, e^{iTx_s} \tilde{g}_r\,, \quad \frac{1}{T}\sqrt{\frac{A - \tilde{a}_{rs}}{2}}\, e^{-iTx_s} \tilde{g}_r\,,$$

$$\frac{1}{T}\sqrt{\frac{A + b_{rs}}{2}}\, e^{jTx_s} g_r\,, \quad \frac{1}{T}\sqrt{\frac{A - b_{rs}}{2}}\, e^{-jTx_s} g_r\,,$$

$$\frac{1}{T}\sqrt{\frac{A + c_{rs}}{2}}\, e^{kTx_s} g_r \,, \qquad \frac{1}{T}\sqrt{\frac{A - c_{rs}}{2}}\, e^{-kTx_s} g_r \,, \; h \,.$$

It is easily checked that these functions have the required properties.

//

9. The Universal Connection for Families

Suppose we have a principal G-bundle over a manifold X, and a family of connections parameterized by some manifold Y. By this we mean that we have:

(i) a fibration $\pi: Z \to Y$ with fiber X

(ii) a principal G-bundle $\bar{\pi}: P \to Z$

(iii) a connection on each $P|_{X_y}$, depending smoothly on $y \in Y$. Here X_y is $\pi^{-1}(y)$.

Using a partition of unity, it is not hard to show that such a family of connections may be extended to a connection on P. By this we mean that there exists a connection on $\bar{\pi}: P \to Z$ such that the set of horizontal vectors for this connection intersected with $\ker \bar{\pi}_* \pi_*$ is precisely the set of horizontal vectors for the original connections.

This connection on $\bar{\pi}: P \to Z$ may be obtained from the universal connection by a map

$$f: Z \longrightarrow BG \ .$$

Thus we have a family of maps

$$f_y: X_y \longrightarrow BG \ , \quad y \in Y \ .$$

These depend smoothly on Y and are connection preserving for the original family of connections. This proves the existence of a universal connection for families.

As an example, let ω_t be a one-parameter family of connections on a principal G-bundle over X. Let ω be the universal connection over BG and let Y be $\mathbb{R}$ and Z be $X \times \mathbb{R}$. Then by the above theorem, there exists a one-parameter family of maps

$$f_t : X \longrightarrow BG$$

such that $f_t^* \omega = \omega_t$.

References

[1] Chern, S. S., Lectures on Exterior Differential Systems. University of California at Berkeley, 1979.

[2] Narasimhan, M. S., Ramanan, S., Existence of universal connections. Amer. J. Math. 83 (1961) 563-572.

[3] Schläfli, L., Nota alla memorici del Sig. Beltrami Sugli spazii di curvatura constante. Ann. di Matem. (2) 5 (1871-1873) 170-193.

[4] Schlafly, R., Universal connections (to appear in Invent. math.).

[5] Spivak, M., A Comprehensive Introduction to Differential Geometry, volume 5. Boston: Publish or Perish, 1975.

Chapter IV

A Chern Number for Gauge Fields on $\mathbb{R}^4$

by Roger Schlafly

1. Introduction

Let G be a compact Lie group and let $\rho: G \to U(N)$ be a representation. Let F be the curvature of some connection on some principal G-bundle over S^4. The Chern number of the associated vector bundle is

$$c_2 = \frac{1}{8\pi^2} \int_{S^4} \operatorname{Tr} \rho(F)^2 .$$

According to the theory of characteristic classes, this quantity is always an integer.

Now consider the curvature F of a connection on a principal G-bundle over $\mathbb{R}^4$. In this paper, we prove

Theorem 1.1. If

$$|F| \leq \frac{c}{r^2 \log r}, \quad r \geq 2$$

for some constant c, then

$$\frac{1}{8\pi^2} \int_{\mathbb{R}^4} \operatorname{Tr} \rho(F)^2$$

is an integer.

This formula is of interest in quantum field theory because the curvature F is the same as a Yang-Mills (gauge) field over Euclidean space. For discussions of related matters, see [1], [2] and [11].

If G is abelian or $N = 1$, the integer obtained in Theorem 1.1 must be zero. If $G = SU(N)$ $(N > 1)$, any integer value is possible.

It is not known whether the hypothesis of Theorem 1.1 can be relaxed to the energy $\int_{\mathbb{R}^4} |F|^2$ being finite, although this is suggested by [2], [6], and [9]. However, it is shown in section 6 that $\frac{1}{8\pi^2}\int_{\mathbb{R}^4} \rho(F)^2$ may not be an integer if we assume only that $|F| \leq c/r^2$. This example is equivalent to the one given in [9].

Theorem 1.1 is proved in sections 2, 3, 4, and 5. Section 3 gives a holonomy formula similar to the one in [10] and may be read independently of the rest of the paper. Section 7 shows how to handle dimensions other than four.

This paper was motivated by a question raised by I. M. Singer. I would like to thank him for many interesting conversations.

2. Outline of the Proof

For the proof of Theorem 1.1 it suffices to consider the associated $U(N)$ bundle. Hence we may, without loss of generality, take $G = U(N)$ and suppress mention of ρ.

Relative to some trivialization of the bundle, the connection A is a one-form on $\mathbb{R}^4$ with values in $u(N)$. The curvature

$$F = dA + A^2$$

is a two-form on $\mathbb{R}^4$ with values in $u(N)$. The Chern-Weil formalism [4, p. 114] tells us that

$$\operatorname{Tr} F^2 = d \operatorname{Tr}(AF - \tfrac{1}{3}A^3) .$$

Let S^3_r be the sphere of radius r in $\mathbb{R}^4$. If $|F| \leq \frac{c}{r^2 \log r}$. Then $\operatorname{Tr} F^2$ is integrable on $\mathbb{R}^4$. By Stoke's theorem,

$$\frac{1}{8\pi^2}\int_{\mathbb{R}^4} \operatorname{Tr} F^2 = \lim_{r\to\infty} \frac{1}{8\pi^2}\int_{S^3_r} \operatorname{Tr}(AF - \tfrac{1}{3}A^3) \ .$$

Given $\beta > 0$ and r sufficiently large, we will show that there exists a smooth map $\tilde{T}\colon S^3_r \to U(N)$ such that

$$\left|\frac{1}{8\pi^2}\int_{S^3_r} \operatorname{Tr}[AF - \tfrac{1}{3}A^3 - \tfrac{1}{3}(\tilde{T}^{-1}d\tilde{T})^3]\right| < \frac{\beta}{2} \ .$$

If we choose r large enough such that

$$\left|\frac{1}{8\pi^2}\int_{\mathbb{R}^4} \operatorname{Tr} F^2 - \frac{1}{8\pi^2}\int_{S^3_r} \operatorname{Tr}(AF - \tfrac{1}{3}A^3)\right| < \frac{\beta}{2} \ ,$$

then it will follow that

$$\left|\frac{1}{8\pi^2}\int_{\mathbb{R}^4} \operatorname{Tr} F^2 - \frac{1}{24\pi^2}\int_{S^3_r} \operatorname{Tr}(\tilde{T}^{-1}d\tilde{T})^3\right| < \beta \ .$$

We now recall a special case of Bott's work on periodicity in K-theory [3].

Theorem 2.1. For $N \geq 2$ we have an isomorphism $\pi_3 U(N) \cong \mathbb{Z}$ given by assigning to a smooth map $\tilde{T}\colon S^3 \to U(N)$ the integer

$$\frac{1}{24\pi^2}\int_{S^3} \operatorname{Tr}(\tilde{T}^{-1}d\tilde{T})^3 \ .$$

Thus for each $\beta > 0$ there exists an integer n such that

$$\left|\frac{1}{8\pi^2}\int_{\mathbb{R}^4} \operatorname{Tr} F^2 - n\right| < \beta \ .$$

Hence $\frac{1}{8\pi^2}\int_{\mathbb{R}^4} \mathrm{Tr}\, F^2$ must be an integer.

3. The Holonomy Propagator

Let M be the sector in $\mathbb{R}^2$ described by polar coordinates (r,θ) with $0 \leq r \leq r_0$ and $0 \leq \theta \leq \theta_0$. Suppose we have a connection A on the trivial principal G-bundle. For this paper we can assume that G is $U(N)$, although Theorems 3.1 and 3.2 hold for any Lie group. A is a one-form on M with values in $u(N)$. The curvature is $F = dA + A^2$.

For $x \in M$ let $c_x\colon [0,1] \to M$ be the curve

$$c_x(t) = tx\ , \quad 0 \leq t \leq 1\ .$$

A section $u\colon M \to \mathbb{C}^N$ of the associated vector bundle is parallel along c_x iff it satisfies

$$c_x^*(du + Au) = 0\ .$$

The fundamental solution to this differential equation is the path ordered exponential

$$P_{exp} - \int_{c_x} A\ .$$

If u is parallel along c_x then

$$u(x) = P_{exp} - \int_{c_x} A\cdot u(0)\ .$$

$P_{exp} - \int_{c_x} A$ is called the holonomy from 0 to x along c_x. It may also be defined as a product integral [8, p. 15].

Theorem 3.1. If

$$T(x) = P_{exp} - \int_{c_x} A$$

then

$$A(x) = -dT \cdot T^{-1} + T\left(\int_{c_x} T^{-1} F T\right) T^{-1} .$$

Proof. Plugging in the vector $\partial/\partial r$, we get

$$A\left(\frac{\partial}{\partial r}\right) = -\frac{\partial T}{\partial r} \cdot T^{-1}$$

which follows from the definition of T. The vector $\partial/\partial\theta$ gives

$$A_2 = -\frac{\partial T}{\partial \theta} \cdot T^{-1} + T\left(\int_0^r T^{-1} F_0 T \, dr\right) T^{-1}$$

where

$$A = A_1 dr + A_2 d\theta$$
$$F = F_0 \, dr \wedge d\theta .$$

To prove this, we first observe that

$$-\frac{\partial A_1}{\partial \theta} = \frac{\partial}{\partial \theta}\left(\frac{\partial T}{\partial r} \cdot T^{-1}\right) = \frac{\partial^2 T}{\partial \theta \partial r} \cdot T^{-1} - \frac{\partial T}{\partial r} T^{-1} \frac{\partial T}{\partial \theta} T^{-1} .$$

Hence

$$\begin{aligned} \frac{\partial}{\partial r}\left[T^{-1}\left(A_2 + \frac{\partial T}{\partial \theta} T^{-1}\right) T\right] &= -T^{-1} \frac{\partial T}{\partial r} T^{-1} A_2 T + T^{-1} \frac{\partial A_2}{\partial r} T + T^{-1} A_2 \frac{\partial T}{\partial r} \\ &\quad - T^{-1} \frac{\partial T}{\partial r} T^{-1} \frac{\partial T}{\partial \theta} + T^{-1} \frac{\partial^2 T}{\partial r \partial \theta} \\ &= T^{-1} A_1 A_2 T + T^{-1} \frac{\partial A_2}{\partial r} T - T^{-1} A_2 A_1 T - T^{-1} \frac{\partial A_1}{\partial \theta} T \\ &= T^{-1} F_0 T . \end{aligned}$$

Continuity of A and $dT \cdot T^{-1}$ at 0 require that

$$\lim_{r \to 0} A_2 = 0$$
$$\lim_{r \to 0} \frac{\partial T}{\partial \theta} \cdot T^{-1} = 0$$

so

$$T^{-1}(A_2 + \frac{\partial T}{\partial \theta} T^{-1})T = \int_0^r T^{-1} F_0 T \, dr$$

which gives the formula we wanted. //

Theorem 3.2. The holonomy from 0 to 0 along ∂M is given by the θ-ordered exponential

$$P_{exp} - \int_M T^{-1} F T \ .$$

Proof. Let

$$V(r,\theta) = P_{exp} \int_0^\theta \int_0^r T^{-1} F_0 T \, dr d\theta$$

with the ordering over θ, so

$$V(r_0,\theta_0) = P_{exp} - \int_M T^{-1} F T \ .$$

By Theorem 3.1,

$$V = P_{exp} - \int_0^{\theta_0} (T^{-1} A_2 T + T^{-1} \frac{\partial T}{\partial \theta}) d\theta \ .$$

We compute

$$\begin{aligned} \frac{\partial}{\partial \theta}(TV)(TV)^{-1} &= T \frac{\partial V}{\partial \theta} V^{-1} T^{-1} + \frac{\partial T}{\partial \theta} T^{-1} \\ &= -T(T^{-1} A_2 T + T^{-1} \frac{\partial T}{\partial \theta}) T^{-1} + \frac{\partial T}{\partial \theta} T^{-1} \\ &= -A_2 \end{aligned}$$

so by the uniqueness of solutions to this ordinary differential equation, there must be some $W: M \longrightarrow G$ independent of θ such that

$$TV = P_{exp} - \int_0^\theta A_2 d\theta \cdot W \ .$$

Writing $T = T(r,\theta)$ and letting $\theta = 0$ we find that $W = T(r,0)$ so

$$T(r,\theta)V(r,\theta) = P_{exp} - \int_0^\theta A_2 d\theta \cdot T(r,0) .$$

Setting $r = r_0$ and $\theta = \theta_0$ gives

$$V(r_0,\theta_0) = T(r_0,\theta_0)^{-1} P_{exp} - \int_0^{\theta_0} A_2 d\theta \cdot T(r_0,0)$$

which is precisely the holonomy from 0 to 0 along ∂M. //

Proposition 3.3. For any one-form B and any curve c,

(i) $|P_{exp}\int_c B| \leq \exp \int_c |B|$

(ii) $|1 - P_{exp}\int_c B| \leq \int_c |B| \exp \int_c |B|$

Proof. Straightforward. //

Corollary 3.4. If H is the holonomy along ∂M then

$$|1-H| \leq \int_M |F| \cdot \exp \int_M |F| .$$

4. Estimates in the Radial Gauge

We now choose a gauge in which there is no holonomy in the radial direction. This means that $A(\partial/\partial r) = 0$ where r is the distance to the origin in $\mathbb{R}^4$. Define $G: \mathbb{R}^4 \to U(N)$ by

$$G(x) = P_{exp} - \int_{c_x} A$$

where

$$c_x(t) = tx , \quad 0 \leq t \leq 1 .$$

Proposition 4.1. (i) Replacing A by $G^{-1}AG + G^{-1}dG$ (and F by $G^{-1}FG$) gives us a radial gauge.

(ii) In this gauge,

$$A(x) = \int_{c_x} F \ .$$

(iii) If c is a constant such that

$$|F| \leq \frac{c}{r^2 \log r}$$

for $r \geq 2$ and

$$|A| \leq \frac{c}{e} \log \log 3$$

for $r = e = 2.718...$, then for $r \geq 3$ we have

$$|A| \leq 2c \frac{\log \log r}{r} \ .$$

Proof. (i) and (ii) follow from Theorem 3.1. For (iii), write $F = F_0 \, dr \wedge d\theta$ so

$$A = \left(\int_0^r F_0 dr \right) d\theta \ .$$

Since $|d\theta| = \frac{1}{r}$ and $|F| = |F_0| \cdot \frac{1}{r}$, we have

$$\begin{aligned} |A| &\leq \left| \int_0^e F_0 dr \right| \cdot |d\theta| + \left| \int_e^r F_0 dr \right| \cdot |d\theta| \\ &\leq \frac{c}{r} \log \log 3 + \frac{1}{r} \int_e^r \frac{c}{r^2 \log r} \, r dr \\ &= \frac{c}{r} \log \log 3 + \frac{c}{r} \cdot \log \log r \\ &\leq 2c \frac{\log \log r}{r} \ . \end{aligned}$$

//

Proposition 4.2. For $r \geq 3$,

$$\left| \int_{S_r^3} \mathrm{Tr}\ AF \right| \leq 15Nc^2 \cdot \frac{\log\log r}{\log r} .$$

Proof. The volume of S_r^3 is $2\pi r^3$, so

$$\begin{aligned} \left| \int_{S_r^3} \mathrm{Tr}\ AF \right| &\leq \int_{S_r^3} |\mathrm{Tr}\ AF| \\ &\leq N \int_{S_r^3} |A| \cdot |F| \\ &\leq N \cdot 2\pi r^3 \cdot 2c\, \frac{\log\log r}{r} \cdot \frac{c}{r^2 \log r} \\ &= 4\pi Nc^2 \cdot \frac{\log\log r}{\log r} . \end{aligned}$$

//

Proposition 4.3. Suppose $r \geq 3$. If

$$|A-B| \leq \frac{\pi c}{r \log r} ,$$

then

$$\left| \int_{S_r^3} \mathrm{Tr}\ A^3 - \int_{S_r^3} \mathrm{Tr}\ B^3 \right| \leq 3000Nc^2 \cdot \frac{(\log\log r)^2}{\log r} .$$

Proof. We have

$$\begin{aligned} |B| &\leq |B-A| + |A| \\ &\leq \frac{\pi c}{r \log r} + 2c\, \frac{\log\log r}{r} \\ &\leq 6c\, \frac{\log\log r}{r} \end{aligned}$$

and

$$A^3 - B^3 = A^2(A-B) + A(A-B)B + (A-B)B^2$$

so

$$|A^3-B^3| \leq \frac{\pi c}{r^2 \log r} \cdot 3(6r \frac{\log\log r}{r})^2$$
$$= \frac{108\pi c^2 (\log\log r)^2}{r^3 \log r} .$$

Thus

$$\left| \int_{S_r^3} \mathrm{Tr}\, A^3 - \int_{S_r^3} \mathrm{Tr}\, B^3 \right| \leq N \int_{S_r^3} |A^3-B^3|$$
$$\leq N \cdot 2\pi r^3 \cdot \frac{108\pi c^2 (\log\log r)^2}{r^3 \log r}$$
$$\leq 300 N c^2 \cdot \frac{(\log\log r)^2}{\log r} . \qquad //$$

5. The Holonomy at Infinity

In this section, we restrict attention to S_r^3 for sufficiently large r. $C_1, C_2, \ldots, C_{13}$ will be constants independent of r.

Fix some "north pole" $p \in S_r^3$, and some great circle Γ from p to $-p$. If $x \in S_r^3$ is not on the "equator" S_r^2, let c_x be the shorter great arc from $\pm p$ to x. (There is a unique great circle in S_r^3 passing through $\pm p$ and x if $x \neq \pm p$.) For x in the "northern hemisphere," let $T(x)$ be the holonomy along c_x. For x in the "southern hemisphere", let $T(x)$ be the holonomy along Γ concatenated with c_x. T is discontinuous on the equator S_r^2.

Lemma 5.1. There exists a constant c_1 such that if T_1, T_2 are the two limiting values of T at a point x in S_r^2, then

$$|T_1-T_2| \leq \frac{c_1}{\log r} .$$

Proof. Choose a surface S in S^3_r having area $\leq 2\pi r^2$ and boundary the union of Γ with the great semicircle from p to $-p$ through x. Then

$$\int_S |F| \leq 2\pi r^2 \cdot \frac{c}{r^2 \log r} = \frac{2\pi c}{\log r} .$$

The holonomy around ∂S is $T_1 T_2^{-1}$, so by Corollary 3.4,

$$|T_1 - T_2| = |1 - T_1 T_2^{-1}| \leq \int_S |F| \exp \int_S |F| .$$

We now choose c_1 so that for large r,

$$\frac{2\pi c}{\log r} \exp\left(\frac{2\pi c}{\log r}\right) \leq \frac{c_1}{\log r} . \qquad //$$

Let $t: S^3_r \to \mathbb{R}$ be the distance to p along S^3_r. Let $\theta = (\theta^1, \theta^2)$ be local coordinates on S^2_r, extended to S^3_r by requiring θ to be constant along each c_x.

Lemma 5.2. For any $x \in S^3_r$,

$$\left| \int_{c_x} T^{-1} F T \right| \leq \frac{2c}{r \log r} .$$

Proof. We suppose that x is on S^2_r, that c_x is taken to be the great arc from p to x, and that $d\theta^1$, $d\theta^2$ are orthonormal at x. The general case will follow easily. Note that

$$|d\theta^i| = \csc\left(\frac{t}{r}\right) , \quad |dt| = 1 .$$

Let

$$F = F_{01}\, dt \wedge d\theta^1 + F_{02}\, dt \wedge d\theta^2 + F_{12}\, d\theta^1 \wedge d\theta^2 ,$$

so

$$\int_{c_x} T^{-1}FT = \sum_{i=1}^{2} \left(\int_{c_x} T^{-1}F_{0i}T \, dt \right) d\theta^i \ .$$

Thus

$$\left| \int_{c_x} T^{-1}FT \right| \leq \sum_{i=1}^{2} \int_{c_x} |F_{0i}| dt$$

$$\leq 2 \int_0^{\pi r/2} |F| \sin\left(\frac{t}{r}\right) dt$$

$$\leq 2 \cdot \frac{c}{r^2 \log r} \cdot r \ . \qquad //$$

If x is a point in the northern hemisphere (i.e. $t < \pi r/2$), then we have by Theorem 3.1,

$$A(x) = -dT \cdot T^{-1} + T\left(\int_{c_x} T^{-1}FT \right) T^{-1} \ . \qquad (5.3)$$

If T_0 is constant, then substituting TT_0 for T leaves (5.3) unchanged. It follows that (5.3) is also valid if x is in the southern hemisphere. Hence we have

Corollary 5.4. (i) $|A + dT \cdot T^{-1}| \leq \frac{2c}{r \log r}$.

(ii) $|dT| \leq c_2 \frac{\log \log r}{r}$ for some constant c_2.

Given $\delta \in (0,1)$, let R be the set of points in S_r^3 with $\frac{\pi r}{2} - \delta < t < \frac{\pi r}{2} + \delta$, and let

$$U(\theta) = \lim_{t \to \frac{\pi r}{2}-} T(t,\theta) \ .$$

From Corollary 5.4(ii), it follows that there exists a constant c_3 such that the inequality

$$|dU| \leq c_3 \frac{\log\log r}{r} \tag{5.5}$$

holds on R. By Lemma 5.1, we can choose δ sufficiently small that on R,

$$|U-T| \leq \frac{2c_1}{\log r} .$$

We suppose that $2c_1/\log r < \frac{1}{2}$ so that $f: R \to u(N)$ may be defined by the power series

$$f = \log U^{-1}T = -\sum_{n=1}^{\infty} \frac{1}{n}(1-U^{-1}T)^n .$$

Then

$$\begin{aligned} |f| &\leq \sum_{n=1}^{\infty} \frac{1}{n}|1-U^{-1}T|^n \\ &\leq \sum_{n=1}^{\infty} \left(\frac{2c_1}{\log r}\right)^n \\ &\leq \frac{4c_1}{\log r} . \end{aligned} \tag{5.6}$$

Let $\phi: \mathbb{R} \to [0,1]$ be a smooth function satisfying

$$\phi(t) = \begin{cases} 1 & |t| \geq 1 \\ 0 & |t| \leq \frac{1}{2} . \end{cases}$$

Define $\tilde{T}: R \to U(N)$ by

$$\tilde{T}(t,\theta) = U(\theta)\exp\left[\phi\left(\frac{t-\frac{\pi r}{2}}{\delta}\right)f(t,\theta)\right]$$

and extend $\tilde{T}$ to be a smooth function on S^3_r by setting it equal to T on $S^3_r - R$.

Lemma 5.7. Let V and E be matrix valued functions.

(i) If $V = e^E$ then $|dV| \le e^{|E|}|dE|$

(ii) If $|1-V| \le \mu < 1$ and $E = \log V$, then

$$|dE| \le \frac{1}{1-\mu}|dV| .$$

Proof. This follows from differentiating the power series for exp and log. //

Lemma 5.8. There exist constants c_4, c_6, c_7 such that on R,

(i) $|df| \le c_4 \frac{\log\log r}{r}$

(ii) $\left|\frac{\partial\tilde{T}}{\partial t}dt\right| \le |d\tilde{T}| \le \frac{c_6}{\delta \log r}$

(iii) $\left|\sum_{i=1}^{2} \frac{\partial\tilde{T}}{\partial\theta^i} d\theta^i\right| \le c_7 \frac{\log\log r}{r}$

Proof. (i) Apply Lemma 5.7(ii) with $\mu = \frac{1}{2}$.

(ii) By Lemma 5.7(i),

$$|d\tilde{T}| \le |dU|e^{|f|} + e^{|f|}|df| + e^{|f|}\left|d[\phi(\frac{t-\frac{\pi r}{2}}{\delta})]\right|\cdot|f| .$$

Then, using (5.5), (5.6), and Lemma 5.8(i),

$$|d\tilde{T}| \le e^{4c_1/\log r}\left(c_3\frac{\log\log r}{r} + c_4\frac{\log\log r}{r} + \frac{c_5}{\delta}\cdot\frac{4c_1}{\log r}\right)$$
$$\le \frac{c_6}{\delta \log r} .$$

(iii) Similar, except that the term involving δ is absent. //

Lemma 5.9. For some c_9,

$$\left|\int_{S_r^3} \mathrm{tr}(T^{-1}dT)^3 - \int_{S_r^3} \mathrm{tr}(\tilde{T}^{-1}d\tilde{T})^3\right| \le c_9 \frac{(\log\log r)^2}{\log r} .$$

Proof.

$$\left|\int_{S^3_r} \mathrm{tr}(T^{-1}dT)^3 - \int_{S^3_r} \mathrm{tr}(\tilde{T}^{-1}d\tilde{T})^3\right| = \left|\int_R \mathrm{tr}(T^{-1}dT)^3 - \int_R \mathrm{tr}(\tilde{T}^{-1}d\tilde{T})^3\right|$$
$$\leq N\int_R |dT|^3 + \int_R |\mathrm{tr}(\tilde{T}^{-1}d\tilde{T})^3| \ .$$

By Lemma 5.8(ii) and 5.8(iii), there exists c_8 such that

$$|\mathrm{tr}(\tilde{T}^{-1}d\tilde{T})^3| \leq c_8 \cdot \frac{1}{\delta \log r} \cdot \left(\frac{\log\log r}{r}\right)^2 \ .$$

The volume of R is bounded by $4\pi r^2\delta$, so from Corollary 5.4(ii) and the above,

$$\left|\int_{S^3_r} \mathrm{tr}(T^{-1}dT)^3 - \int_{S^3_r} \mathrm{tr}(\tilde{T}^{-1}d\tilde{T})^3\right|$$
$$\leq N\cdot 4\pi r^2\delta\cdot\left(c_2\frac{\log\log r}{r}\right)^3 + 4\pi r^2\delta\cdot c_8\cdot\frac{1}{\delta\log r}\cdot\left(\frac{\log\log r}{r}\right)^2$$
$$\leq c_9\,\frac{(\log\log r)^2}{\log r} \ . \qquad //$$

We now complete the proof of Theorem 1.1. From $\mathrm{tr}\,F^2 = d\,\mathrm{tr}(AF - \frac{1}{3}A^3)$ and

$$|F| \leq \frac{c}{r^2\log r} \ ,$$

it follows that

$$\left|\frac{1}{8\pi^2}\int_{\mathbf{R}^4} \mathrm{tr}\,F^2 - \frac{1}{8\pi^2}\int_{S^3_r} \mathrm{tr}(AF - \frac{1}{3}A^3)\right| \leq \frac{N}{8\pi^2}\int_r^\infty \left(\frac{c}{\rho^2\log\rho}\right)^2 2\pi\rho^3 d\rho = \frac{c_{10}}{\log r} \ .$$

From Proposition 4.2,

$$\left|\frac{1}{8\pi^2}\int_{\mathbf{R}^4} \mathrm{tr}\,F^2 + \frac{1}{24\pi^2}\int_{S^3_r} \mathrm{tr}\,A^3\right| \leq c_{11}\,\frac{\log\log r}{\log r} \ .$$

Corollary 5.4(i) shows that the hypothesis of Proposition 4.3 is satisfied if $B = -dT \cdot T^{-1}$, so

$$\left| \frac{1}{8\pi^2} \int_{\mathbb{R}^4} \operatorname{tr} F^2 - \frac{1}{24\pi^2} \int_{S^3_r} \operatorname{tr}(T^{-1} dT)^3 \right| \leq c_{12} \frac{(\log \log r)^2}{\log r} .$$

By Lemma 5.9,

$$\left| \frac{1}{8\pi^2} \int_{\mathbb{R}^4} \operatorname{tr} F^2 - n \right| \leq c_{13} \frac{(\log \log r)^2}{\log r} ,$$

where n is the integer

$$\frac{1}{24\pi^2} \int_{S^3_r} \operatorname{tr}(\tilde{T}^{-1} d\tilde{T})^3 .$$

The proof of Theorem 1.1 is now finished by letting r tend to infinity.

6. An Example

Let $T: S^3 \to SU(2)$ be the standard identification. Let ω be the pull-back to $\mathbb{R}^4$ of $T^{-1}dT$ by the radial projection. The structural equation is

$$d\omega + \omega^2 = 0 .$$

Let $f: [0,\infty) \to \mathbb{R}$ be a smooth function satisfying

$$f(r) = \begin{cases} 0 & 0 \leq r \leq \frac{1}{2} \\ a & r \geq 1 \end{cases}$$

for some real number a. Using r for the radial coordinate, $f(r)\omega$ is a well-defined smooth form on $\mathbb{R}^4$, and we define it to be a connection on the trivial $SU(2)$ bundle.

We compute

$$F = f'dr \wedge \omega + (f^2-f)\omega^2$$

so $F \in O(1/r^2)$. Also

$$F^2 = 2(f^2-f)f'dr \wedge \omega^3$$

has compact support and

$$\begin{aligned} \frac{1}{8\pi^2}\int_{\mathbb{R}^4} \operatorname{Tr} F^2 &= \frac{1}{8\pi^2}\cdot\int_0^\infty 2f'(r)(f^2(r)-f(r))dr\cdot\int_{S^3} \operatorname{Tr} \omega^3 \\ &= \frac{1}{8\pi^2}\cdot\left[\frac{2}{3}f^3(r) - f^2(r)\right]_0^\infty \cdot 12 \text{ volume } S^3 \\ &= 2a^3 - 3a^2 . \end{aligned}$$

By varying a, we may obtain any real number.

7. A Generalization

Theorem 1.1 generalizes to the following situation. Let G be a compact Lie group and let $\rho: G \to U(N)$ be a representation. Let F be the curvature of some connection on some principal G-bundle over $\mathbb{R}^{2n}$. Let

$$C_n = \frac{(-1)^{n+1}}{n} \operatorname{Tr}\left(\frac{\rho(F)}{2\pi i}\right)^n .$$

If the connection extends to a connection on some bundle over S^{2n}, then the Chern number [7] of the associated vector bundle is $\int_{\mathbb{R}^{2n}} C_n$. This is an integer and is always divisible by $(n-1)!$ (see [4, p. 77] or [5, p. 156]).

Theorem 7.1. Suppose $n > 1$. If there exists a constant C such that

$$|F| \leq \frac{C}{r^2 \log r}, \quad r \geq 2 ,$$

then

$$\frac{1}{(n-1)!} \int_{\mathbb{R}^{2n}} C_n$$

is an integer. Any integer value is possible if $G = SU(N)$ and $N \geq n$.

Proof. The proof requires only minor modifications of the proof of Theorem 1.1, which we now discuss. According to the Chern-Weil theory [4, p. 114],

$$\operatorname{Tr} F^n = nd\int_0^1 \operatorname{Tr}\{A[tF + (t^2-t)A^2]^{n-1}\}dt .$$

It follows as before that in the radial gauge,

$$\int_{\mathbb{R}^{2n}} \operatorname{Tr} F^n = n\int_0^1 (t^2-t)^{n-1}dt \cdot \lim_{r\to\infty} \int_{S_r^{2n-1}} \operatorname{Tr} A^{2n-1} .$$

An elementary integration by parts gives

$$\int_0^1 t^{n-1}(1-t)^{n-1}dt = \frac{[(n-1)!]^2}{(2n-1)!} .$$

Approximating A by $-d\tilde{T}\cdot\tilde{T}^{-1}$ for some smooth $\tilde{T}: S^3_r \to U(N)$, we find that

$$\frac{1}{(n-1)!}\int_{\mathbb{R}^{2n}} C_n$$

is approximated by

$$\frac{-1}{(2\pi i)^n}\cdot\frac{(n-1)!}{(2n-1)!}\int_{S^{2n-1}_r} \mathrm{Tr}(\tilde{T}^{-1}d\tilde{T})^{2n-1} .$$

This is an integer because it is the integral that gives Bott periodicity [3],

$$\pi_{2n-1}U(N) \cong \mathbb{Z} ,$$

for $n \leq N$. The integral is zero if $n > N$. //

The proof of Theorem 7.1 breaks down if $n = 1$. In order to estimate the holonomy around a closed curve near infinity, we expressed the curve as the boundary of a surface where the curvature was small and used Corollary 3.4. But the sphere of radius r in $\mathbb{R}^{2n}$ is simply-connected only if $n > 1$.

The following theorem shows that Theorem 7.1 actually fails if $n = 1$.

Theorem 7.2. Let F be a two-form on $\mathbb{R}^2$ with values in some Lie algebra. Then F is the curvature of some connection on a principal bundle over $\mathbb{R}^2$.

Proof. Write

$$F = f(x,y)\, dx \wedge dy .$$

Then F is the curvature of the connection

$$A(x,y) = \left(\int_0^x f(x',y)dx'\right)dy \ . \qquad //$$

References

1. Atiyah, M. F., Hitchin, N. J., Singer, I. M., Self-duality in four-dimensional Riemannian geometry, Proc. R. Soc. Lond. A 362 (1978) 425-461.

2. Belavin, A. A., Polyakov, A. M., Schwartz, A. S., Tyupkin, Yu. S., Pseudoparticle solutions of the Yang-Nills equations, Phys. Lett. 59B (1975) 85-87.

3. Bott, R., Seeley, R., Some remarks on the paper of Callias, Comm. Math. Phys. 62 (1978) 235-245.

4. Chern, S. S., Complex Manifolds without Potential Theory, 2nd ed., New York: Springer Verlag, 1979.

5. Griffiths, P., Adams, J., Topics in Algebraic and Analytic Geometry, Princeton, N.J.: Princeton University Press, 1974.

6. Marino, E. C., Swieca, J. A., Non-integer topological charge and infiniteness of action, P. U. C., Nota Cientifica 7/78 (1978).

7. Milnor, J., Stasheff, J., Characteristic Classes, Princeton, N.J.: Princeton University Press, 1974.

8. Nelson, E., Topics in Dynamics, I: Flows, Princeton, N.J.: Princeton University Press, 1970.

9. Schaposnik, F. A., Solomin, J. E., Gauge field singularities and noninteger topological charge, J. Math. Phys. 20 (1979) 2110-2114.

10. Schlesinger, L., Parallelverschiebung und Krümmingstensor, Math. Ann. 99 (1928) 413-434.

11. Uhlenbeck, K. K., Removable singularities of Yang-Mills fields, Bull. Amer. Math. Soc. (new ser.) 1 (1979) 579-581.

Chapter V

Convergence of an Index Problem

In [2], Atiyah, Patodi, and Singer give a formula for the index of a particular type of boundary value problem. Applying their formula to the Dirac operator on the ball of radius r in $\mathbb{R}^4$, we consider the limit as r tends to infinity.

1. The Dirac Operator

Let M be a closed riemannian spin manifold, and let $E \to M$ be a hermitian vector bundle with connection. Let $V \to M$ be the bundle of spinors. Then we can construct (see [1]) the Dirac operator with coefficients in E:

$$A: V \otimes E \longrightarrow V \otimes E \ .$$

The curvature $F \in \Lambda^2(T^*M) \otimes \mathrm{Hom}(E,E)$ of E may be written

$$F(x) = \sum_{i<j} F_{ij}(x)\ e_i \wedge e_j \ , \quad x \in M$$

where $F_{ij}(x) \in \mathrm{Hom}(E_x, E_x)$ and $\{e_i\}$ is an orthonormal base for T_x^*M. We put a pointwise norm on $\Lambda^2(T^*M) \otimes \mathrm{Hom}(E,E)$ by

$$|F(x)| = \sum_{i<j} |F_{ij}(x)|$$

where we use the usual operator norm on $\mathrm{Hom}(E,E)$. Define

$$C(F): V \otimes E \longrightarrow V \otimes E$$

by

$$C(F)(v \otimes e) = \sum_{i<j} e_i \cdot e_j \cdot v \otimes F_{ij}(e)$$

where the dot indicates clifford multiplication. Then

$|C(F)| \leq |F|$. A computation (see [1], [3], or [4]) gives:

Proposition 1.1. $A^2 = \nabla^*\nabla + C(F) + \frac{1}{4}R$ where ∇ is the covariant derivative and R is the scalar curvature of M.

Corollary 1.2. If $|F| < \frac{1}{4}R$ then $\ker A = 0$.

Proof. If $A\psi = 0$ then

$$\begin{aligned} 0 &= \langle \psi, A^2\psi \rangle \\ &= \langle \psi, (\nabla^*\nabla + C(F) + \tfrac{1}{4}R)\psi \rangle \\ &= \langle \nabla\psi, \nabla\psi \rangle + \langle \psi, (C(F) + \tfrac{1}{4}R)\psi \rangle \ . \end{aligned}$$

But the last line is positive unless $\psi = 0$. //

2. The Eta Invariant

Let X be the unit ball in $\mathbb{R}^4$, with some riemannian metric. Let E be a hermitian vector bundle over X with connection ω and curvature F. Let D be the Dirac operator on X with coefficients in E, going from $V_+ \otimes E$ to $V_- \otimes E$. Let A be the Dirac operator on ∂X with coefficients in E (restricted to ∂X). Let P be the projection onto the eigenspaces of A corresponding to nonnegative eigenvalue. Let

$$\begin{aligned} \text{index } D = &\dim\{\psi: D\psi = 0 \text{ and } P(\psi|_{\partial X}) = 0\} \\ &- \dim\{\psi: D^*\psi = 0 \text{ and } (1-P)(\psi|_{\partial X}) = 0\} \end{aligned}$$

Let p_1 be the Chern-Weil curvature form of X. Let n be $\dim \ker A$. Let $\{\lambda\}$ be the eigenvalues of A. Let

$$\eta(s) = \sum_{\lambda} (\text{sign } \lambda)|\lambda|^{-s} \ , \quad \text{Re } s > 4$$

and we analytically continue to the whole complex plane.

Theorem 2.1 [2]. If the metric and connection are a product near the boundary, then

$$\text{index } D = \frac{1}{8\pi^2}\int \text{tr } F^2 - \frac{1}{24}\int_X p_1 - \frac{1}{2}h - \frac{1}{2}\eta(0) \ .$$

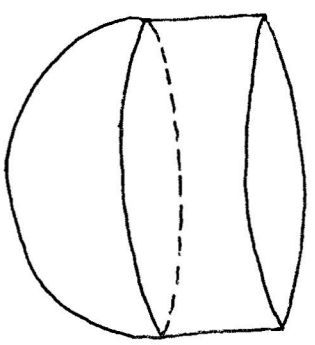
To obtain a formula valid without the "product near the boundary" hypothesis, one can use the following fact asserted in [2]: index D will be unchanged by a perturbation of the metric and connection that leaves the restrictions to the boundary fixed.

Since a bundle over X must be topologically trivial, we can use the Chern-Weil formalism to write

$$\int_X \text{tr } F^2 = \int_{\partial X} \text{tr}(\omega\, d\omega + \frac{2}{3}\omega^3) \ .$$

Hence this quantity is independent of the connection on the interior. Thus we can drop the assumption on the connection in Theorem 2.1.

Now suppose X is the ball of radius r in $\mathbb{R}^4$ with its standard flat metric. Obviously p_1 is zero. It is possible to deform this metric to one that is a product near the boundary and still has p_1 identically zero; see the figure:



Thus, if E is an arbitrary vector bundle with connection over X, we have

$$\text{index } D = \frac{1}{8\pi^2}\int_X \text{tr } F^2 - \frac{1}{2}h - \frac{1}{2}\eta(0) \tag{2.2}$$

We now apply the vanishing theorem of section one. The scalar curvature of ∂X is $R = 6/r^2$. Suppose that for some r_0 we have

$$|F| < \frac{3}{2r^2}, \quad r \geq r_0 .$$

Then $h = 0$.

In general, $\frac{1}{2}\eta(0)$ depends continuously on the metric and connection except for integer jumps which are caused by an eigenvalue of A crossing the origin. If $r \geq r_0$, this does not happen, so $\frac{1}{2}\eta(0)$ depends continuously on r. Thus we have:

Theorem 2.3. Suppose that $\int_{\mathbb{R}^4} \text{tr } F^2$ exists and that for some r_0, $|F| < \frac{3}{2r^2}$ if $r \geq r_0$. Then each term in (2.2) converges as r tends to infinity.

3. Further Remarks

Let E be a hermitian vector bundle with connection over $\mathbb{R}^4$. Let F be its curvature. If E extends to S^4, then

$$\frac{1}{8\pi^2}\int_{\mathbb{R}^4} \text{tr } F^2 \tag{3.1}$$

is an integer because it is the second Chern number. It is also the index of the Dirac operator with coefficients in E.

If E does not extend to S^4, one can still hope that (3.1) is an integer if F obeys suitable decay conditions at infinity. One

approach to this question would be to try to realize (3.1) as the index of D, the Dirac operator with coefficients in E. On $\mathbb{R}^4$, this operator is not Fredholm, so we restrict it to the ball of radius r and let r tend to infinity. We must impose the boundary conditions of section two because there is an obstruction to the existence of elliptic boundary conditions of the classical type, and the obstruction is nonzero in this case.

Proposition 3.2. If F has compact support, then

$$\lim_{r\to\infty} \text{index } D = \frac{1}{8\pi^2}\int_{\mathbb{R}^4} \text{tr } F^2 \ .$$

Proof. By the vanishing theorem, $h = 0$ for large r. Also, S^3 has an orientation reversing isometry, so $\eta(0) = 0$. The result follows from (2.2). //

Obviously, we would like to weaken the hypotheses of this proposition. Suppose we only assume that $\int_{\mathbb{R}^4} \text{tr } F^2$ exists and that $|F| < \frac{\epsilon}{r^2}$ for some small $\epsilon > 0$. The example in section six of [5] can be arranged to satisfy $|F| < \frac{\epsilon}{r^2}$ for all r, so index D is independent of r and

$$\text{index } D = \lim_{r\to 0} \text{index } D = 0 \ .$$

Hence equation (2.2) gives

$$\frac{1}{8\pi^2}\int_{\mathbb{R}^4} \text{tr } F^2 = \lim_{r\to\infty} \frac{1}{2}\eta(0) \ .$$

According to [5], the left hand side may not be an integer.

The moral here is that we want to assume that F decays at least as fast as $o(1/r^2)$. This is consistent with [5] where it is shown that this is precisely the condition necessary for there to be very little holonomy around closed loops near infinity.

Now suppose that $|F| \leq \frac{\text{const}}{r^2 \log r}$ for large r. Then it is shown in [5] that $\frac{1}{8\pi^2}\int_{\mathbb{R}^4} \operatorname{tr} F^2$ is an integer. Thus $\lim_{r\to\infty} \frac{1}{2}\eta(0)$ is an integer. It would be very surprising if this could be nonzero so we are led to the following conjecture: If $|F| \leq \frac{\text{const}}{r^2 \log r}$ then

$$\lim_{r\to\infty} \text{index } D = \frac{1}{8\pi^2}\int_{\mathbb{R}^4} \operatorname{tr} F^2 \ .$$

[1] Atiyah, M.F., Hitchin, N.J., Singer, I.M., Self-duality in four-dimensional Riemannian geometry, Proc. R. Soc. Lond. A 362 (1978) 425-461.

[2] Atiyah, M.F., Patodi, V.K., Singer, I.M., Spectral asymmetry and Riemannian geometry. I., Math. Proc. Camb. Soc. 77 (1975) 43-69.

[3] Lichnerowicz, A., Spineurs harmoniques, C.R. Hebd. Séanc. Acad. Sci., Paris A-B 257 (1963) 7-9.

[4] Michelsohn, M.L., Clifford and spinor cohomology of Kähler manifolds, preprint.

[5] Schlafly, R., A Chern number for gauge fields over $\mathbb{R}^4$.